Science of the Total Environment 905 (2023) 166992

Contents lists available at ScienceDirect

Science of the Total Environment

journal homepage: www.elsevier.com/locate/scitotenv

Optimization of biogas production from straw wastes by different pretreatments: Progress, challenges, and prospects

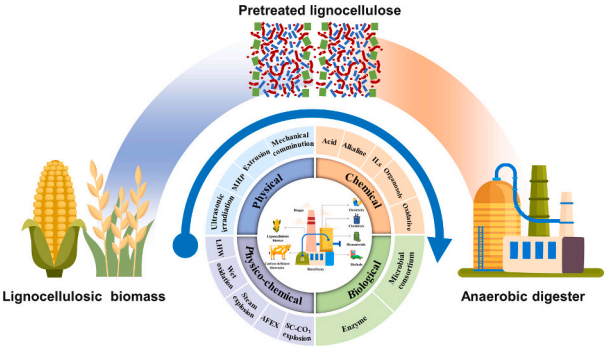
Jing Wang, Dongmei Ma, Yu Lou, Jun Ma, Defeng Xing *

State Key Laboratory of Urban Water Resource and Environment, School of Environment, Harbin Institute of Technology, Harbin 150090, China

HIGHLIGHTS

- Newly developed pretreatment methods for lignocellulosic biomass are supplemented.
- Physical-chemical pretreatment is more flexible and effective for straw wastes.
- Artificial enzymes development and enzyme immobilization expand biological method.
- Economic-environmental effectiveness evaluation is a priority for pretreatments.
- Micro-aeration, electro-fermentation, and conductive materials fortify straw AD.

GRAPHICAL ABSTRACT



ARTICLE INFO

Editor: Qilin Wang

Keywords:
Pretreatment
Anaerobic digestion (AD)
Lignocellulosic biomass
Crop straw
Biogas

ABSTRACT

Lignocellulosic biomass (LCB) presents a promising feedstock for carbon management due to enormous potential for achieving carbon neutrality and delivering substantial environmental and economic benefit. Bioenergy derived from LCB accounts for about 10.3 % of the global total energy supply. The generation of bioenergy through anaerobic digestion (AD) in combination with carbon capture and storage, particularly for methane production, provides a cost-effective solution to mitigate greenhouse gas emissions, while concurrently facilitating bioenergy production and the recovery of high-value products during LCB conversion. However, the inherent recalcitrant polymer crystal structure of lignocellulose impedes the accessibility of anaerobic bacteria, necessitating lignocellulosic residue pretreatment before AD or microbial chain elongation. This paper seeks to explore recent advances in pretreatment methods for LCB biogas production, including pulsed electric field (PEF), electron beam irradiation (EBI), freezing-thawing pretreatment, microaerobic pretreatment, and nanomaterials-based pretreatment, and provide a comprehensive overview of the performance, benefits, and drawbacks of the traditional and improved treatment methods. In particular, physical-chemical pretreatment emerges as a flexible and effective option for methane production from straw wastes. The burgeoning field of nanomaterials has provoked progress in the development of artificial enzyme mimetics and enzyme immobilization techniques, compensating for the intrinsic defect of natural enzyme. However, various complex factors, such as economic effectiveness, environmental impact, and operational feasibility, influence the implementation of LCB pretreatment processes. Techno-economic analysis (TEA), life cycle assessment (LCA), and artificial intelligence technologies provide efficient means for evaluating and selecting pretreatment methods. This paper

* Corresponding author at: School of Environment, 2614, 73 Huanghe Road, Nangang District, Harbin, Heilongjiang Province 150090, China.
E-mail address: dxing@hit.edu.cn (D. Xing).

https://doi.org/10.1016/j.scitotenv.2023.166992
Received 27 June 2023; Received in revised form 9 September 2023; Accepted 9 September 2023
Available online 17 September 2023
0048-9697/© 2023 Elsevier B.V. All rights reserved.

addresses current issues and development priorities for the achievement of the appropriate and sustainable utilization of LCB in light of evolving economic and environmentally friendly social development demands, thereby providing theoretical basis and technical guidance for improving LCB biogas production of AD systems.

1. Introduction

Given the exhaustion of fossil fuel resources and increasing environmental concerns, energy security and global energy consumption have become burning issues for the international community (Abraham et al., 2020). Increasing awareness of climate change and the growing demand for energy have dramatically boosted the development of renewable low-carbon energy (Wang et al., 2023). LCB, an abundant and low-cost renewable resource, is perceived as capable of carbon neutral and negative with the combination of carbon capture and storage schemes. LCB is an umbrella term for dry phytomass. The annual global yield of LCB is about 200 billion tons, and the bioenergy captured by terrestrial plants per annum is about 3–4 times higher than the energy demands of humans (Guo et al., 2015; Shrivastava and Sharma, 2022). The traditional treatments of LCB are incorporation and incineration, which can result in serious land waste, air pollution, and other problems (Wang et al., 2021b). In consequence, the proper utilization of LCB to recovery biofuels and bioenergy is of great significance for alleviating the global energy crisis and greenhouse gas emissions (Fig. 1). Significantly, LCB is expected to provide about 38 % of the global direct fuel supply as well as 17 % of global electricity by 2050 (Shrivastava and Sharma, 2022).

Anaerobic digestion (AD) is known to be one of the most potent and cost-effective ways for LCB utilization, integrating economic and environmental benefits (Sun et al., 2023). LCB contains three primary components of cellulose, hemicellulose, and lignin, which generally account for 35–50 %, 20–35 %, and 10–25 %, respectively (Rodrigues et al., 2022). These components interact to form a recalcitrant polymer crystal structure that impedes the hydrolysis of lignocellulose (Deivayanai et al., 2022). In this case, the pretreatment of lignocellulose is a critical step in the decomposition of lignocellulosic natural refractory structure and the release of carbohydrates during biomass conversion (Bhatia et al., 2020). Increasing the specific surface area of feedstock is deemed an effective way for enzyme access and microbial attachment to promote cellulose degradation (Yang et al., 2015a). Many pretreatment methods have proven to change the surface area of LCB, such as grinding, wetting, drying, etc., thereby significantly affecting biogas production (Ma et al., 2022). Likewise, hemicellulose and lignin removal is important for increasing the surface area of cellulose in cellulose-hemicellulose-lignin matrix, thus enabling the cellulose more accessibility and susceptibility to enzymes. Hemicellulose can be effectively removed by multiple pretreatments such as dilute acid as well as steam explosion (Periyasamy et al., 2022). Some inhibitors (furfural, hydroxymethylfurfural, phenolic compounds, etc.) or negative environmental impacts may occur during the processing procedure, thus limiting the practical application of these pretreatments.

Lignin is the most recalcitrant obstacle to lignocellulose decomposition. Pretreatments of wet oxidation, organosolv, alkaline, and biological processes have been shown to be effective in delignification (Millati et al., 2020; Yang et al., 2015a). The cellulose decrystallization is a pivotal step in the utilization of feedstock, which can generally be realized by pretreatment based on numerous inorganic acids. Research on LCB pretreatment methods is developing rapidly (Fig. 2). Cold isostatic pressure pretreatment has been confirmed to destroy the microstructure of rice straw, enhance the accessibility of microbes, and further facilitate the methane production during AD process by 76 % as compared to the untreated group (Yang et al., 2023b). Interestingly, microaerobic pretreatment was reported to be an effective means to assist microbial pretreatment of cellulosic straw to enhance AD performance, and specifically, the methane production increased by 88.7 % from microaerobic pretreated rice straw with straw decomposing consortia addition (Amin et al., 2021). Keep abreast of new developments on LCB pretreatment for biogas production will provide theoretical and technical guidance for the industrialization and large-scale application of the pretreatment process in actual AD plants. From the perspective of economic, environmental, and social sustainability, meanwhile,

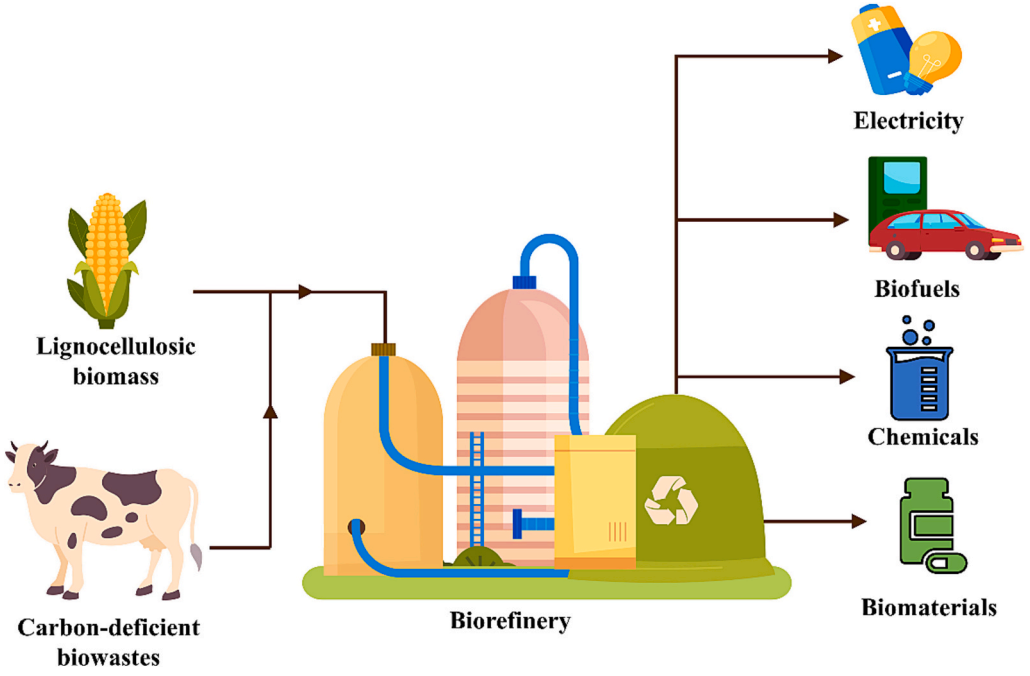


Fig. 1. The biorefinery operations of lignocellulosic biomass.

pretreatment cost accounts for about 30 % of the total cost of biofuels, and its selection is closely associated with the type of feedstock and end product (Alonso et al., 2013). The selection of LCB pretreatment and process condition optimization will observably affect regional, and even global biogas market and climate change, making it of great significance to evaluate the techno-economic and environmental benefits of pretreatment strategies.

To better understand the basic principles and technical features of various pretreatments, this review outlines the current status of traditional physical, chemical, and biological pretreatment methods, and introduces the new development of emerging approaches based on PEF, EBI, freezing-thawing, microaerobic, and nanomaterials-based pretreatments available for LCB biogas production (Table 1). Moreover, the future research priorities for the evaluation of pretreatment economic effectiveness and environmental impact by techno-economic analysis (TEA) coupled with life cycle assessment (LCA) of LCB are summarized and discussed in this paper.

2. Physical pretreatment

Compared with chemical and biological pretreatments, physical pretreatment can alter the physical properties, rather than the chemical composition, of feedstock (Gu et al., 2015; Zheng and Rehmann, 2014). There are few potential fermentation inhibitors generated during the pretreatment process, which is one of the obvious advantages of physical pretreatment (Karunanithy and Muthukumarappan, 2011). Current research primarily focuses on mechanical comminution, extrusion, microwave heating pretreatment, and ultrasonic irradiation. Table 2 summarizes the relevant studies on the improvement of the production of biogas from LCB by physical pretreatment.

2.1. Mechanical comminution

Mechanical comminution, including grinding, milling, chipping, etc., can facilitate the digestibility of lignocellulose by reducing particle size and fiber crystallinity and improving mass transfer. Grinding and milling are the most frequently used practices for lignocellulose pretreatment; they have been reported to shorten the acclimation time and fermentation period and promote the methane production of LCB (Zheng and Rehmann, 2014). For example, comminution pretreatment was reported to shorten the fermentation period and enhance the methane production from maize silage and maize straw by 38.4 % and 8.3 %, respectively (Witaszek et al., 2020). In addition, the reduction of particle size could benefit the AD performance of rice straw by enhancing cellulose degradation instead of acting on hemicellulose or lignin, and the methane yield increased from 107 mL/g VS to 197 mL/g VS as the straw particle size reduced from 20 mm to 0.075 mm (Dai et al., 2019). The selection of grinding and milling relies largely on the moisture content of biowaste (Neshat et al., 2017). After chipping or milling/grinding, the particle size of wastes is respectively about 10–30 or 0.2–2 mm (Sun and Cheng, 2002). Nonetheless, an excessively small particle size may lead to volatile fatty acids (VFAs) accumulation, which is detrimental to methanogenesis during AD (Izumi et al., 2010). Mechanical comminution is an expensive process with most of the input energy wasted as heat energy. Further, comminution pretreatment has little effect on the delignification of lignocellulose (Shikinaka et al., 2018). The combination of mechanical comminution and other pretreatment methods may be more beneficial for the energy recovery of lignocellulosic residues.

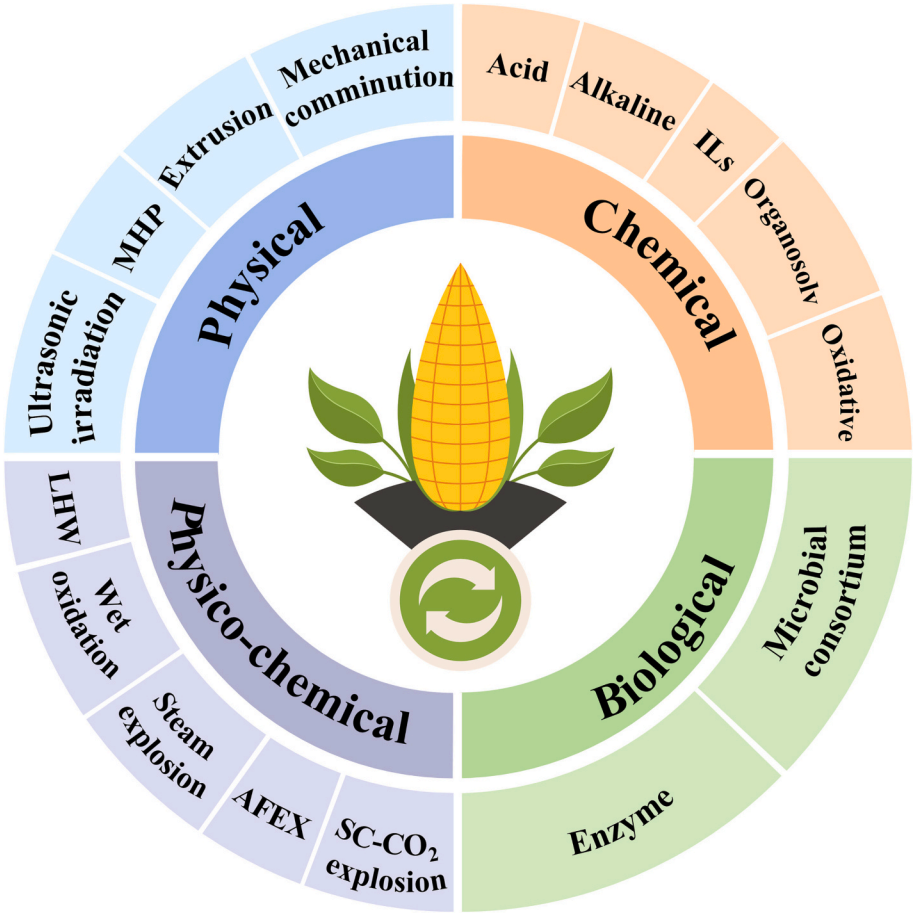


Fig. 2. The various pretreatment methods for lignocellulosic biomass.

Table 1
Comparison on the merits and current limitations of different pretreatment methods.

Classification	Specific method	Merits	Current limitations	Ref.
Physical pretreatment	Mechanical comminution	· No production of ethanol fermentation inhibitors	· Little effect on the delignification · High energy requirement	(Shikinaka et al., 2018)
	Extrusion	· Simplicity of operator; · Eco-friendly	· High energy consumption and high cost · Formation of inhibitors	(Zheng et al., 2014)
	Microwave heating pretreatment	· Eco-friendly · Effectively destroy the structure of lignocellulose · Rapidly heat a large volume and reduce the treatment time	· High installation cost · Formation of inhibitors	(Yu et al., 2019; Zheng et al., 2014)
	Ultrasonic irradiation	· Eco-friendly · Effectively disrupt cell wall structure, increase the specific surface area, and reduce the degree of polymerization	· Expensive and difficult in industrial application	(Yu et al., 2019; Zheng et al., 2014)
Chemical pretreatment	Acid pretreatment	· Effectively dissolve a portion of hemicellulose	· Excessive loss of cellulose and produce inhibitors · Equipment corrosion · Environmental toxicity · Expensive operation and maintenance costs	(Alvira et al., 2010; Rabemanolontsoa and Saka, 2016)
	Alkaline pretreatment	· Effectively disintegrate lignin and reduce crystallinity, and improve the subsequent hydrolysis and biodegradability of lignocellulose	· Environmental toxicity · Expensive operation and maintenance costs	(Du et al., 2019; Liu et al., 2015)
	Ionic liquid pretreatment	· Low energy consumption, simple operation, and environmental protection · Effectively remove lignin and hemicellulose and reduce cellulose crystallinity	· Expensive and difficult to recycle and reuse · Incomplete decomposition of hemicellulose and lignin	(Li et al., 2010; Ravindran and Jaiswal, 2016)
	Organosolv pretreatment	· Effectively dissolve and extract lignin from lignocellulosic structures · High-purity cellulose recovery	· High reagent cost, capital investment, and energy consumption	(Zhou et al., 2018)
	Oxidative pretreatment	· Effectively destroy the structures of hemicellulose and lignin	· High investment · Unnecessary loss of cellulose and hemicellulose · Formation of inhibitors	(Croce et al., 2016; Zheng et al., 2014)
Physico-chemical pretreatment	Liquid hot water pretreatment	· Effectively dissolve hemicellulose and disintegrate lignin · Little corrosion effect on equipment · Low amount of pollutants, low cost in capital investments, need less energy and chemical inputs, and generate minimal inhibitory products and waste	· Difficult optimization of reaction conditions · Dilute liquid streams leading to significant impacts on fermentation efficiency	(Chen et al., 2022)
	Wet oxidation	· Effectively dissolve hemicellulose and lignin	· Loss of organic matter such as hemicellulose · Formation of inhibitors, such as furfural and soluble aromatic compounds	(Zhou et al., 2023)
	Steam explosion	· Cost-effective, efficient, and environmentally friendly technique · Low energy input · Less water per biomass weight · Effectively depolymerize and extract soluble hemicellulose	· Degradation of hemicellulosic sugars · Incomplete disintegration of the structure linking lignin and carbohydrates · Formation of inhibitors	(Hoang et al., 2023)
	Ammonia fiber expansion	· Almost no significant formation of inhibitors · Effective on herbaceous crops with lower lignin content · Low cost · Pretreated LCB can be readily hydrolyzed and fermented without any water washing or additional detoxification	· No significant lignin or hemicellulose removal · Less effective on woody biomass	(Zhao et al., 2020)
	Supercritical carbon dioxide explosion	· Less corrosiveness · Clean, low-cost, and has high diffusivity · No significant formation of inhibitors	· High installation cost and investment	(Dharmaraja et al., 2023)
Biological pretreatment	Microbial consortium-based pretreatment	· Far lower energy input and no chemicals · Much milder environmental conditions so as to few inhibitors	· Long pretreatment period · Not as efficient as chemical pretreatments	(Cheng and Whang, 2022)
	Enzyme-based pretreatment	· Far lower energy input and no chemicals · Much milder environmental conditions so as to few inhibitors · Short reaction time · Relatively low sugar loss	· Not as efficient as chemical pretreatments · High economic cost · and environmental sensitivity	(Cheng and Whang, 2022)
Emerging pretreatment strategies	Pulsed electric field	· Non-thermal and energy-efficient technique · More efficient energetic utilization for LCB · Short reaction time and effectively shorten retention times of subsequent AD process · No requirement of chemicals · Environmentally-friendly and safe for the operation	· Lack of reliable and viable industrial equipment · High initial investment costs · High power supply requirement	(Kovačić et al., 2021)
	Electron beam irradiation	· Simple operation · Short processing time	· Incomplete decomposition of lignocellulose	(Fei et al., 2020)

(*continued on next page*)

Table 1 (*continued*)

Classification	Specific method	Merits	Current limitations	Ref.
		· Large processing capacity · Easy control · No chemical requirements · No pollution to the environment		
	Freezing-thawing pretreatment	· No additional energy input · Environmentally friendly technique	· Highly affected by natural environment · Incomplete decomposition of lignocellulose	(Yang et al., 2023a)
	Microaerobic pretreatment	· Simple operation · Low energy input · High economic and environmental benefits	· Unclear mechanism	(Cai et al., 2021)
	Nanomaterials-based pretreatment	· Possess both chemical catalysts and biocatalyst properties · Higher catalytic stability, ease of modification and lower manufacturing cost than protein enzymes · High recoverability and reusability	· Unclear mechanism · Difficult in rational design and surface chemistry	(Jiang et al., 2019)

Table 2
Physical pretreatment methods for the enhancement of biogas/methane production from LCB.

Physical pretreatment	Pretreatment condition	Biomass	AD condition	Results	Ref.
Comminution	Particle size: 0.25–1.25 mm;	Maize silage	Batch and wet digestion	38.4 % increase in methane yield	(Witaszek et al., 2020)
	Particle size: ≥0.25 mm;	Maize straw silage	Batch and wet digestion	8.3 % increase in methane yield	(Witaszek et al., 2020)
	Particle size: 20, 1, 0.15, 0.075 mm;	Rice straw	Batch, 37 °C, 25 d	50.5 %–84.1 % increase in methane yield	(Dai et al., 2019)
	Particle size: <0.45 mm;	Rice straw	Batch, 35 °C, 40 d	About 50 % increase in methane yield	(Zhang et al., 2015)
	Particle size: 0.28–50 mm;	Rice straw	Batch,37 °C, 60–90 d	30.0 % increase in methane yield	(Chen et al., 2014)
	3 h dry milling and 6 h wet milling;	*Pennisetum* hybrid	Batch, 35 °C, 46 d	41 % and 24 % increase in methane yield	(Dong et al., 2017)
	Particle size: 33–6 mm;	Sorghum hybrid; forage rye; triticale;	Batch, 25 °C, 90 d	11 %–13 % increase in methane yield	(Herrmann et al., 2012)
Extrusion	Temperature: 150 °C and 175 °C; Pretreatment time: 58, 62, 70, 72 s;	Maize silage; Maize straw silage;	Batch and wet digestion	6.4 %–12.4 % increase in methane yield	(Witaszek et al., 2020)
	Particle size: <2.5 mm;	Rice straw	Batch, 37 °C, 60–90 d	72.2 % increase in methane yield	(Chen et al., 2014)
	No details	Vegetable, fruit, and flower market wastes	Batch, 37 °C, 30 d	31.0 % increase in methane yield	(Mozhiarasi et al., 2020)
	Particle size: <0.45 mm;	Rice straw	Batch, 35 °C, 40 d	About 51 % increase in methane yield	(Zhang et al., 2015)
Microwave heating	Specific energy 11,670; 26,130; and 42,000 kJ/kg TS;	Bamboo shoot shell	Batch, 37 °C, 45 d	13.2 %, 21.0 %, and 30.1 % increase in methane yield	(Fang et al., 2020)
	Power: 400–1600 W;	Wheat straw	Batch, 35 °C, 60 d	20.2 %–27.5 % increase in methane yield	(Jackowiak et al., 2011)
	Power: 500 W; Temperature: 100 °C;	Hyacinth	Batch, 35 °C, 50–60 d	38.3 % increase in methane yield	(Zhao et al., 2017a)
Ultrasonic irradiation	Specific energy: 0–8064 kJ/kg;	Cattle manure and straw wheat mixture	Batch, 35 °C, 20 d	59.6–64.2 % increase in biogas yield	(Zieliński et al., 2017)
	Power intensity: 0, 189.39, 284.09, 378.79 kJ; Pretreatment time: 0, 20, 30, 40 min;	Maize straw	Batch, 35 °C, 42 d	27.7 %–69.7 % increase in methane yield	(Zou et al., 2016)
	Power: 200 W, 450 W, 700 W; Pretreatment time: 2, 10 min;	Olive pomace	Batch, 35 °C	0.9 %–13.5 % increase in methane yield	(Elalami et al., 2020)
	Frequency: 20 kHz; Pretreatment time: 10 min;	Wheat straw	Batch, 37 °C, 50 d	17.0 % increase in biogas yield	(Korai et al., 2020)

2.2. Extrusion

Extrusion is an eco-friendly and simple pretreatment method, integrating multifarious operations in a continuous process (Akobi et al., 2016; Pérez-Rodríguez et al., 2017b). The feedstock is input from one side of the extrusion machine and conveyed along the barrel length *via* a driving screw, during which the substrates are subjected to frictional heating, mixing, and intense shearing (Hjorth et al., 2011). The pressure released during extrusion contributes to the considerable abrasion of materials and consequently decomposes the structural components of cells. Moreover, extrusion pretreatment can depolymerize the lignocellulose and protein (Hjorth et al., 2011; Zheng and Rehmann, 2014). Extrusion has extensive application prospects and exhibits commendable flexibility and adaptability for process amplification and modification (Pérez-Rodríguez et al., 2017a). The extrusion pretreatment of fruit wastes has been found to improve methane production by 31 % (Mozhiarasi et al., 2020), and extruded rice straw has yielded a 72.2 % increase in methane production (Chen et al., 2014). Unlike fruit wastes and rice straw, a mere 0–15 % increase in methane production from extruded maize silage has been observed (Panepinto and Genon, 2016).

Significantly, the positive effect of extrusion on methanogenesis is very much dictated by substrate type and operating conditions; therefore, extrusion pretreatment is generally adopted to assist with other pretreatment approaches to achieve better fermentation performance (Gu et al., 2015; Pérez-Rodríguez et al., 2017b; Sun et al., 2020; Zhang et al., 2012). For instance, sequentially extruded and enzyme-pretreated vine-trimming shoots have been found to yield 40 % more methane production than untreated raw materials (Pérez-Rodríguez et al., 2017b). Combined extrusion and sodium hydroxide (NaOH) pretreatment (3.0 % NaOH, 35 °C, 48 h) has been found to achieve a 54.0 % increase in methane production compared with untreated rice straw (Zhang et al., 2015). More economical methods must be developed to reduce the high energy consumption and cost of extrusion pretreatment.

2.3. Microwave heating pretreatment

Microwave heating pretreatment (MHP) is an effective substitute for traditional heating, and is recognized as a "green" technology without the addition of any chemical reagents (Aguilar-Reynosa et al., 2017). The rapid and uniform heating of MHP can dramatically reduce the energy requirement and heating period of materials (Motasemi and Ani, 2012; Sapci, 2013). MHP can effectively alter cellulose ultrastructure, partially remove hemicellulose as well as lignin components, and disrupt wax layer on cellulosic biomass surface, making cellulose more accessible to microorganisms and enzymes (Periyasamy et al., 2022). The enhancement of MHP pretreatment on corn straw AD has been confirmed, and the methane yield increased by 73.08 % after MHP pretreatment for 10 min at 150 °C (Wang et al., 2021a). In comparison with untreated biomass, the cumulative methane production of MHP-pretreated bamboo shoot shells with targeted specific energies of 11,670, 26,130, and 42,000 kJ/kg TS was found to be increased by 13.2 %, 21.0 %, and 30.1 %, respectively (Fang et al., 2020). The microwave pretreatment of hyacinth has been found to yield a 38.3 % increase in cumulative methane production compared with water-heating pretreatment at the same temperature (Zhao et al., 2017a). Although microwave pretreatment has been found to speed up the reaction rate of AD, no biogas production improvement was observed *via* microwave-pretreatment switchgrass assay (Jackowiak et al., 2011). Similarly, in another study, the methanogenesis of several crop straws (barley, wheat, and oat) was not improved after microwave radiation at 200 or 300 °C, and the biogas production decreased with the increase of the temperature (Sapci, 2013). The reason for this might be the formation of some heat-induced inhibitors, e.g., phenolic compounds or furfural (Jing and LÜ, 2007; Ximenes et al., 2010). Consequently, the control of the pretreatment conditions is critical to the reduction of the formation of these inhibitors. Currently, microwave radiation is normally assisted with chemical or biological pretreatment at relatively low temperatures (Fang et al., 2020; Sun et al., 2020).

2.4. Ultrasonic irradiation

Ultrasonic action is caused by acoustic cavitation rather than direct interactions with molecules. The application of ultrasonic irradiation to lignocellulosic materials has revealed that the structure of biomass can be decomposed by cavitation force, thereby improving the hydrolysis efficiency and sugar yield (Karimi et al., 2014; Wang et al., 2017). As waves travel through the medium, regions of expansion and compression form, and the dissolved gas becomes bubbles under the action of external forces. Periodic wave irradiation increases the energy accumulated by the bubble vibration until it implodes, releasing a large amount of local energy and thus increasing diffusion (Martinez-Jimenez et al., 2017). In one study, the methane production of an ultrasonic-pretreated cornstalk and dairy manure mixture increased by 69.7 % with an ultrasonic energy input of 284.1 kJ/kg TS (Zou et al., 2016). Ultrasound irradiation has been reported to substantially increase the biogas production from a cattle manure and wheat straw mixture (Zieliński et al., 2019). Ultrasonic pretreatment is normally used in combination with other pretreatment methods. One study suggested that ultrasonic pretreatment was deleterious to biogas production, while the enzymatic hydrolysis and ultrasonic co-treatment of corncob resulted in a 2.3-fold increase in methane production compared with untreated biomass (Perez-Rodriguez et al., 2016).

3. Chemical pretreatment

Chemical pretreatment can destroy the original physical and chemical structure and properties of lignocellulosic materials *via* the addition of chemical agents. Studies on the improvement of biogas production from agricultural wastes by chemical pretreatment are summarized in Table 3.

3.1. Acid pretreatment

Acid pretreatment can effectively improve AD efficiency of LCB. The glycosidic bonds in hemicellulose and cellulose are easily affected by acids; hence, an appropriate acid pretreatment can dissolve a portion of the hemicellulose in LCB, thus making cellulose more accessible to enzymes (Sun et al., 2016). The pretreatment of lignocellulosic biowaste with concentrated acids (30–70 %) at 40 °C and dilute acids (0.1 %) at 230 °C has been reported (Zheng and Rehmann, 2014). Although concentrated acid pretreatment is rather efficient for the hydrolysis of cellulose, it may cause an excessive loss of cellulose and produce a large number of inhibitors. The problems of equipment corrosion, environmental toxicity, acid recovery, and expensive operation and maintenance costs also make concentrated acid pretreatment less attractive for practical application (Alvira et al., 2010; Rabemanolontsoa and Saka, 2016). On the contrary, dilute acid pretreatment avoids or alleviates the aforementioned drawbacks and is a regularly used chemical pretreatment for LCB (Alvira et al., 2010; Mikulski and Klosowski, 2018). The common dilute acids include hydrochloric acid (HCl), sulfuric acid (H_2SO_4), nitric acid (HNO_3), as well as phosphoric acid, among which H_2SO_4 is the cheapest and most widely applied agent (Kim et al., 2014). In one study, among a series of H_2SO_4 concentrations [1.0 %–6.0 % (v/v)], the 3.0 % (v/v) H_2SO_4 pretreatment of an apple pulp and chicken manure mixture exhibited maximal cumulative methane yield, contributing to a 1.8-fold increase compared with the untreated mixture (Senol, 2020). Three different inorganic acids (HCl, H_2SO_4, HNO_3) were applied to pretreat the marine macroalgae *Padina tetrastromatica* in a previous study, and the results revealed that dilute H_2SO_4 pretreatment contributed to the optimal performance of phenol removal and biohydrogen production (Radha and Murugesan, 2017). In another study, the methane yield of H_2SO_4-pretreated rye straw (121 °C for 1 h) exhibited a 3.2-fold increase as compared to that of thermal-pretreated rye straw under the same conditions (Domanski et al., 2020). Organic acids (oxalic, maleic, fumaric, and succinic acids, etc.) are promising alternatives to inorganic acids for avoiding machine corrosion and reducing the energy requirement for acid recovery. It has been reported that maleic acid is superior to sulfuric acid in cellulose hydrolysis and hemicellulose degradation (Lee and Jeffries, 2011). Organic acid pretreatment is more suitable for lignocellulosic materials with a high cellulose content and a low hemicellulose content, such as aquatic plants (Rabemanolontsoa and Saka, 2016).

3.2. Alkaline pretreatment

Alkaline pretreatment is quite favorable for lignocellulosic bioconversion. Alkaline treatment is able to disintegrate lignin and reduce crystallinity, and thus to improve the subsequent hydrolysis and biodegradability of lignocellulose. The presence of hydroxyl ions in the solution is able to break the hydrogen bonds between cellulose-hemicellulose, saponify the ester bonds between cellulose-lignin, and break the chemical bonds between lignin-carbohydrates (Du et al.,

Table 3
Chemical pretreatment methods for the enhancement of biogas/methane production from LCB.

Chemical pretreatment	Pretreatment conditions	Biomass	AD conditions	Maximum biogas/methane production under specific pretreatment conditions	Ref.
Acid pretreatment	Dilute H_2SO_4 [1 %–10 % (w/v)]; 121 °C; 1–2 h;	Rye straw	Batch, 35 °C, 14 d	3.2-fold increase in methane yield with 10 % H_2SO_4, 1 h	(Domanski et al., 2020)
	Dilute H_2SO_4 [1.0 %–6.0 % (v/v)]; 25 °C; 48 h;	Apple pulp and chicken manure	Batch, 35 °C, 30 d	1.8-fold increase in methane yield with 3.0 % H_2SO_4	(Şenol, 2020)
	Dilute H_2SO_4 [1 %–5 % (v/v)]; 121 °C; 0–1.5 h	Water hyacinth	Batch, 28-30 °C, 90 d	131.45 % increase in biogas production with 5 % H_2SO_4, 1 h	(Sarto et al., 2019)
	Dilute H_2SO_4 [2 %–6 % (v/v)]; 30 °C; 48 h;	*Salvinia molesta*	Batch, 30 °C, 30 d	81.78 % increase in biogas yield with 4 % H_2SO_4, 1 h	(Syaichurrozi et al., 2019)
	Dilute H_2SO_4, CH_3COOH, H_3PO_4 [2 %–8 % (m/m)]; 60 °C; 3 h;	Corn stover	Batch, 55 °C, 25 d	40.75 % increase in biogas yield with 6 % H_3PO_4	(Tian et al., 2016)
Alkaline pretreatment	NaOH, KOH [1 %–5 % (w/w)]; 25-100 °C; 24 h;	Chestnut shell	Batch, 37 °C, 30 d	2.4-fold increase in methane yield with 3 % NaOH, 100 °C	(Bianco et al., 2021)
	NaOH [1 %–6 % (w/w)]; 25 °C; 48 h;	Hazelnut husks	Batch, 39 °C, 30 d	162.2 % increase in methane yield with 4 % NaOH	(Senol, 2020)
	NaOH [1 %–6 % (w/w)]; 20 ± 2 °C; 72 h;	Teff straw	Batch, 35 °C, 50 d	40 % increase in biomethane production with 4 % NaOH	(Chufo et al., 2015)
	NaOH [1.6 % (w/w)]; 30 °C; 24 h;	Wheat straw	Batch, 37 °C, 40 d	15 % increase in biomethane production	(Mancini et al., 2018b)
	KOH [1 %–9 % (w/w)]; Ambient temperature; 72 h;	Wheat straw	Batch, 35 °C, 40 d	45 % increase in biogas production with 6 % KOH	(Jaffar et al., 2016)
	$Ca(OH)_2$ [1.7 %–11.9 % (w/w)]; Room temperature; 96 h;	Sugarcane bagasse	Batch, 37 °C, 35 d	23.3 % increase in biogas production with 8.5 % $Ca(OH)_2$	(Mustafa et al., 2018)
	$Ca(OH)_2$ [5 %–15 % (w/w)]; 25 °C; 72 h;	Rice straw	Batch, 35 °C, 40 d	36.7 % increase in biogas production with 10 % $Ca(OH)_2$	(Gu et al., 2015)
	KOH [0.5 %–3.0 % (w/v)], $Ca(OH)_2$ [0.5 %–3.0 % (w/v)]; 20 °C; 24 h;	Corn stover	Batch, 37 °C, 30 d	77.01 % increase in methane yield with 0.5 % KOH + 2.0 % $Ca(OH)_2$	(Li et al., 2015b)
	liquid ammonia [2 % (w/w)]; −20 and 20 °C; 7 d;	Corn stalk	Batch, 35 °C, 50 d	49.6 % increase in biogas production with −20 °C frozen 12 h, 20 °C thawed 12 h and NH_3 water 7 d	(Li et al., 2020)
	liquid ammonia [2 %,4 %,6 % (w/w)]; 30 ± 2 °C; 168 h;	Corn stover	Batch, 35 °C, 65 d	26.7 % increase in biogas production with 4 % ammonia	(Li et al., 2015a)
Ionic liquids pretreatment	[C2mim]Cl, [C4mim]Cl and [C6mim]Cl; 120 °C; 2–4 h;	Water hyacinth, rice straw, mango leaves and spruce	Batch, 35 °C, 30 d	97.6 %, 40.8 %, 23.4 % and 57.0 % increases in biogas production from water hyacinth, rice straw, mango leaves and spruce with [C4mim]Cl, 2 h	(Gao et al., 2013)
	NMMO [85 % (w/w)]; 120 °C; 3 h;	Wheat straw	Batch, 37 °C, 40 d	11 % increase in biomethane production	(Mancini et al., 2018b)
	NMMO [85 % (w/w)]; 90 °C; 3–30 h;	Barley straw and forest residues	Batch, 55 °C, 45 d	2.1-fold increase in methane production from forest residues with NMMO, 30 h	(Kabir et al., 2014)
	NMMO [85 % (w/w)]; 130 °C; 1,3,15 h;	Rice straw, triticale straw, and softwood	Batch, 55 °C, 42 d	400 %–1200 % increase in methane yield	(Teghammar et al., 2012)
	NMMO [73 %,79 %,85 % (w/w)]; 90-120 °C; 1–5 h;	Oil palm empty fruit bunch	Batch, 55 °C, 50 d	48 % increase in biomethane production	(Purwandari et al., 2013)
Organosolv pretreatment	Ethanol [50 % (v/v)]; 180 °C; 1 h;	Wheat straw	Batch, 37 °C, 40 d	15 % increase in biomethane production	(Mancini et al., 2018b)
	Ethanol [50 % (v/v)]; 150 and 180 °C; 1 h;	Rice straw, hazelnut skin, and cocoa bean shell	Batch, 37 °C, 43 d	42 % increase in biomethane production from rice straw with 50 % ethanol, 180 °C	(Mancini et al., 2018a)
	Isopropanol [50 % (v/v)]; 140, 160, 180, and 200 °C; 30 and 60 min;	Sunflower stalks	Batch, 37 °C, 45 d	112.9 % increase in methane yield with 50 % isopropanol, 180 °C, 60 min	(Hesami et al., 2015)
	Ethanol [1 %,3 %,5 % (w/v)], 100-120 °C, 1 h;	Rice Husk	Batch, 37 °C, 21 d	121.8 % increase in biogas production with 3 % C_2H_5OH, 100 °C	(Olugbemide et al., 2019)
Oxidative pretreatment	H_2O_2 [4 %,8 %,12 % (v/v)]; 36-58 °C; 32.5–87.5 min;	Corn stalk	Batch, 37 °C, 34 d	22 % increase in biogas production with 12 % H_2O_2 (pH 11.5), 58 °C, 58 min	(Venturin et al., 2018)
	H_2O_2 [4 % (w/w)]; 30, 55, 80 °C; 24 h;	Sunflower stalks	Batch, 35 °C, longer than 18 d	33 % increase in biogas production with 4 % H_2O_2, 55 °C	(Monlau et al., 2012)
	PMS 1 mmol/g VS, PF 1 mmol/g VS; Room temperature; 2 h;	Cotton stalk and cow manure mixture	Batch, 35 °C, 57 d	1.3-fold increase in methane yield with PF	(Wang et al., 2021b)

2019). The cleavage of these bonds observably improves the dissolution of hemicellulose as well as lignin, and consequently enlarges the surface area over which cellulose is exposed to enzymes. Alkali reagents such as sodium (NaOH), potassium (KOH), and calcium hydroxides [$Ca(OH)_2$] and urea or ammonia solutions have been extensively used in LCB pretreatment, among which NaOH has been the most widely utilized alkali in recent decades. Compared with untreated teff straw, 4 % NaOH-pretreated straw with a solid loading of 80 g/L has been found to contribute to 40.0 % and 48.1 % increases in biomethane production and the biomethane content, respectively (Chufo et al., 2015). In one investigation, NaOH pretreatment with1.5 % (w/v) and NaOH-assisted photocatalytic pretreatment [1.5 % (w/v) NaOH, 0.25 g/L TiO_2)] respectively resulted in 71 % and 122 % increases in methane production from rice straw (Sabeeh et al., 2020). However, the recovery problems and the potential risks of water pollution and land salinization induced by NaOH have promoted the application of more eco-friendly KOH, which can be utilized as an agricultural fertilizer in favor of soil reclamation (Du et al., 2019; Liu et al., 2015). KOH pretreatment of wheat straw was reported to shorten the fermentation period and improve the methanogenic capacity and fertility value, and 6 % KOH-pretreated wheat straw contributed to a 41 % increase in cumulative biomethane yield as compared to untreated straw (Jaffar et al., 2016).

However, the high cost of KOH restricts its industrial application (Shen et al., 2019). $Ca(OH)_2$ is deemed a cost-effective substitute for KOH, and is 8–10 times cheaper than NaOH (Khor et al., 2015). It has been reported that the methane yield of 8 % $Ca(OH)_2$-pretreated corn straw was 105.3 % higher than that of raw straw (Song et al., 2014). In addition to the aforementioned benefits of promoting biogas production, the residual potassium, magnesium, and calcium ions after pretreatment can be utilized as commendable soil amendments (Jaffar et al., 2016). The alkaline reagents can increase the initial pH value of AD and can also serve as nutrient additives. The application of urea solution or ammonium hydroxide has been reported to facilitate the decomposition of lignocellulose, increase the nitrogen source, and balance the C/N ratio (Oladi and Aita, 2017). By report, the co-treatment of NaOH and urea under low-temperature conditions increased the hydrogen production of rice stalk by 161.9 % compared with the untreated material (Dong et al., 2018). Although NaOH pretreatment of wheat straw can efficiently decompose the lignin structure and enhance the hydrolysis of organics in the initial stage of fermentation, the formation of humic substances inhibits the activity of acetogens and suppresses methanogenesis, and co-digestion with waste activated sludge (WAS) is an effective method for the alleviation of such inhibition (Chen et al., 2020).

3.3. Ionic liquid pretreatment

Ionic liquids (ILs) are environmentally friendly molten organic salts composed of ions that exist in liquid form at room temperature, and are non-combustible, lowly volatile, and easy to recycle. IL pretreatment can effectively remove lignin as well as hemicellulose and reduce cellulose crystallinity without producing degradation products that inhibit enzymes or fermentation microorganisms (Li et al., 2010). IL pretreatment has the benefits of low energy consumption, simple operation, and environmental protection (Ravindran and Jaiswal, 2016). There is growing interest in ILs, which can avoid the toxicity and corrosiveness of inorganic acids (Nargotra et al., 2018). The most commonly used ILs are 1-ethyl-3-methyl imidazolium acetate ([Emim][Ac]), 1-ethyl-3-methyl imidazolium chloride ([Emim][Cl]), 1-butyl-3-methylimidazolium chloride, 1-butyl-3-methylimidazolium dicyanamide, *N*-methylmorpholine-N-oxide (NMMO), cholinium ILs (especially amino acid-based ILs), etc. Compared with untreated straw, the IL pretreatment of [Emim][Ac] and [Emim][Cl] has been found to observably enhance the enzymatic hydrolysis and shorten the hydrolysis period of barley straw (Lara-Serrano et al., 2018). Similarly, 12 different cholinium ILs have been found to benefit enzymatic hydrolysis of rice straw (Xu et al., 2016). In one study, the biogas production of 1-N-butyl-methlyimidazolium chloride-pretreated water hyacinth, rice straw, mango leaves, and spruce increased by 97.6 %, 40.8 %, 23.4 %, and 57.0 %, respectively, as compared with the corresponding raw materials (Gao et al., 2013). Moreover, the biomethane potential of NMMO-pretreated wheat straw (120 °C, 3 h) has been found to be increased by 11 % as compared to untreated straw (Mancini et al., 2018b). The residual hemicellulose and lignin restrain the further decomposition of lignocellulose after IL mono-pretreatment. Research agendas should maintain focus on improving the recovery efficiency of ILs. The combination of ILs and other pretreatments might be a viable strategy to overcome the previously noted drawbacks (Chang et al., 2016; Liu et al., 2018).

3.4. Organosolv pretreatment

Organosolv pretreatment is an efficient method for simultaneous polysaccharide hydrolysis and delignification. One of the major advantages of organosolv pretreatment is that relatively pure lignin can be recovered as a solid material and carbohydrate (i.e., syrup) (Zhao et al., 2009). Organosolv pretreatment is normally conducted by various organic or aqueous-organic solvents (ethanol, acetone, acetic acid, methanol, etc.) to solubilize lignin, thus enabling easier enzymatic hydrolysis for cellulose (Alvira et al., 2010). The ethanol organosolv process is the most frequently utilized organosolv pretreatment in lignocellulose biorefining for biogas and bioethanol production (Amiri et al., 2014). Ethanol organosolv pretreatment at 180 °C for 1 h has been found to have a major impact on the dissolution of hemicellulose, and the cumulative methane production of the pretreated wheat straw was 15 % higher than that of the raw material (Mancini et al., 2018b). Rice husk pretreated with a 3 % concentration of ethanol organosolv at 100 °C for 60 min has been found to achieve a 2.2-fold increase in biogas production compared with the untreated test (Olugbemide et al., 2019). Catalytic organosolv is more effective than sole organosolv for the treatment of LCB, which has positive effects on the enhancement of delignification and cellulose enzymatic digestibility. Acids (e.g., H_2SO_4, HCl, and salicylic acids) and alkalis are the most frequently selected catalysts for catalytic organosolv pretreatment (Zhang et al., 2016). In one investigation, five organosolv pretreatments of diethylene glycol, acetone, methanol, ethanol, and butanol with a catalyst of 23 mol/m^3 H_2SO_4 were found to enhance the straw digestibility, besides, the highest cellulose conversion rate and ethanol yield were observed *via* ethanol pretreatment assay with a pretreatment temperature of 180 °C for 40 min (Salapa et al., 2017). Alkali-catalyzed additives are promising alternatives due to the problems of the inhibitors and equipment corrosion in acid-catalyzed organic solvent pretreatment. Organosolv pretreatment catalyzed by organic amines of lignocellulosic materials is a temperate and effective approach for the bioconversion of lignocellulose into high-yield fermentable sugars, as well as high-quality salt-free lignin (Tang et al., 2017). Satisfactory delignification properties can be obtained by increasing the pretreatment temperature instead of adding chemical agents.

3.5. Oxidative pretreatment

Oxidative pretreatment has been proven to enhance the biological conversion of LCB by effectively destroying the structures of hemicellulose and lignin in lignocellulose (Zheng and Rehmann, 2014). Ozone, a powerful oxidant, is usually selected as a "green" and effective pretreatment method, which is more inclined to facilitate the biodegradability of lignocellulosic residues by disintegrating the lignin content rather than by altering cellulose and hemicellulose. Ozone molecules can break down in water to produce hydroxyl radicals (•OH), which makes a great contribution to the decomposition of refractory substrates. In recent decades, ozone pretreatment has been widely applied to convert LCB into various value-added chemicals and materials, such as fuels and thermal energy (Travaini et al., 2016). In one study, 45-min ozonolysis treatment resulted in 158 % and 166 % enhancements in the accumulative hydrogen yield from wheat and barley stalks, respectively (Wu et al., 2013a; Wu et al., 2013b). Furfural and hydroxymethylfurfural are not detected during the pretreatment process, although the high investment of ozone pretreatment limits its large-scale application.

Hydrogen peroxide (H_2O_2) can effectively decompose hemicellulose and lignin (Paudel et al., 2017). H_2O_2 can self-decompose *in situ* to form •OH, which exhibits a stronger oxidizing property (E^0 = 2.8 V) than H_2O_2 (E^0 = 1.8 V) (Yang et al., 2019). The methane yield of H_2O_2-pretreated (2.7 % w/w TS) rice straw has been found to be increased by 88 % as compared to that of raw straw (Song et al., 2013). H_2O_2 pretreatment is greatly influenced by the pH value of the reaction system, and the optimal pH is about 11.5 (Gould, 1984). Several soluble inhibitors (e.g., aromatic compounds) are inevitably produced during lignin oxidization by H_2O_2 pretreatment (Zheng and Rehmann, 2014). The H_2O_2 concentration is a pivotal factor influencing the AD performance, and the selection of the H_2O_2 concentration requires condition optimization prior to use. Oxidative pretreatment has been reported to be detrimental to biogas production for the non-selectivity of oxidants and free radicals, and may result in an unnecessary loss of cellulose and hemicellulose (Croce et al., 2016). The combination of low-dosage H_2O_2 and other pretreatments, such as NaOH (Yuan et al., 2018), acetic acid (Mota et al.,

2019), phosphoric acid (Qiu et al., 2018), etc., has been demonstrated to be an effective means to overcome the disadvantages. For instance, in one investigation, the proportion of cellulose in alkaline H_2O_2-pretreated corn stalk increased by 73.4 % and the proportion of lignin decreased by 71.6 % as compared with the untreated stalk (Venturin et al., 2018).

To date, many innovative oxidative pretreatment methods are used to improve the production of short-chain fatty acids (SCFAs) and biogas from municipal wastes and lignocellulose materials, such as free nitrous acid (FNA), calcium peroxide (CaO_2), potassium ferrate (PF), and peroxymonosulfate (PMS). FNA is generally used with other pretreatment methods (peracetic acid, PF, freezing process, etc.) to facilitate WAS decomposition and SCFA production (Hu et al., 2020; Liu et al., 2020a; Liu et al., 2020b). By report, CaO_2 enhanced the methane yield of sludge, which increased linearly with the increase in the amount of CaO_2 addition from 0 to 0.14 g/g VSS (Wang et al., 2019). PF and PMS have been proven to facilitate WAS hydrolysis and SCFA production (He et al., 2018; Li et al., 2018; Yang et al., 2019). The previous research of the present group indicated that PF and PMS co-treatment had the highest lignin removal efficiency for the mixture of cow manure-cotton straw, while the maximum cumulative methane yield was observed after PF pretreatment (Wang et al., 2021b). The reason for this might be the loss of biodegradable organic matter due to the non-selective free radicals and the introduction of a high concentration of sulfate. The pretreatment of lignocellulosic wastes by innovative oxidative methods is less studied and in urgent need of more effort.

4. Physico-chemical pretreatment

The combination of physical and chemical pretreatments can compensate for the limitations of individual pretreatment. Physico-chemical pretreatment can not only enhance the destruction of cellulosic biomass and the subsequent AD efficiency, but also reduce the use of chemical reagents and thus to improve its environmental benefits. Researches on the improvement of biogas production from straw wastes *via* physico-chemical pretreatment are shown in Table 4.

4.1. Liquid hot water pretreatment

The use of liquid hot water (LHW) is a hydrothermal pretreatment method that can effectively decompose the recalcitrant polymer structure of LCB by dissolving hemicellulose, disintegrating lignin, thus further promoting the accessibility of cellulose to microbes and enzymes (Zhuang et al., 2016). As the sole solvent for LHW pretreatment, high-temperature liquefied water has little corrosion effect on equipment and almost no inhibition of byproduct generation during the pretreatment process (Alvira et al., 2010). LHW has been widely used to enhance methanogenesis from LCB, including hybrid *Pennisetu* (Kang et al., 2020), wheat straw (Shang et al., 2019), *Eucalyptus grandis* (Yu et al., 2010), hybrid poplar (Kim et al., 2009), sweet sorghum bagasse (Yu et al., 2011), etc. LHW pretreatment at 175 °C for 35 min has been found to contribute to a 32.9 % increase in methane yield of hybrid *Pennisetum* compared with the raw material (Kang et al., 2020). In another work, maximal methane yield was found for straw pretreated by LHW at 175 °C for 30 min, and was 62.9 % higher than that of the raw material (Shang et al., 2019). Therefore, the optimization of LHW pretreatment conditions for specific substrates is quite necessary for the different physicochemical characteristics of biomass.

4.2. Wet oxidation

Wet oxidation is highly effective for hemicellulose and lignin removal from lignocellulose (Zhou et al., 2023). It is realized by the simultaneous addition of air/oxygen and water/H_2O_2 at 125–300 °C and 0.5–20 MPa (Kumar and Sharma, 2017; Zheng and Rehmann, 2014). This process is exothermic, and the released energy can maintain the system temperature at a desirable level, thereby reducing the energy input. The oxygen pressure, temperature, pretreatment time, and water content are pivotal factors affecting wet oxidation pretreatment. When the pretreatment temperature reaches 170 °C, the water will behave as an acid and can catalyze the hydrolysis reaction (Kumar and Sharma, 2017). The presence of oxygen during wet oxidation pretreatment can accelerate the reaction rate, with the effect of oxygen concentration-dependent. Air is generally selected as a cost-effective substitute for pure oxygen due to its low operation cost. Wet oxidation pretreatment has been proven to be a satisfactory method to enhance the production of bioethanol from lignocellulosic residues (Bjerre et al., 1996). In one experiment, oxidation pretreatment (195 °C, 12 bar oxygen, 15 min, 2 g/L Na_2CO_3) was found to enhance the ethanol production of three crop straws (winter rye, oilseed rape, faba bean) and increase the biogas production of winter rye straw (Petersson et al., 2007). In another study, thermo-oxidation pretreatment (4.5 % H_2O_2, 100 °C, 30 min) with 2 % wet NaOH presoaking yielded the highest methane production from a mixture of pretreated soybean straw and farm wastewater, which increased by 62 % as compared to the untreated straw (Xiong et al., 2020). Wet oxidative pretreatment is also a non-selective process characterized by the loss of organic matter. The high cost of H_2O_2 and the flammability of pure oxygen limit the industrial application of wet oxidation (Kumar and Sharma, 2017).

4.3. Steam explosion

Steam explosion is defined as autohydrolysis, i.e., externally-provided, high-pressure, saturated steam can rapidly heat biomass pellets, which is followed by a sharp depressurization to terminate the reactions and thus decompose the feedstocks (Hoang et al., 2023). The temperature and pressure ranges commonly used for steam explosion are 160–250 °C and 5–50 bar, respectively (Abraham et al., 2020). As previously reported, the temperature (140–220 °C) and pressure (2–15 min) of steam explosion were set as gradients to explore the optimal operating conditions, and the maximum biogas yield was observed in steam-exploded cornstalk at 160 °C for 2 min, which was 25 % higher than that in untreated cornstalk (Lizasoain et al., 2017). In another experiment, steam explosion pretreatment effectively increased biogas production, and coffee husk pretreated at 120 °C for 60 min exhibited maximal methane yield and electricity production (Baeta et al., 2017). The reactions of different components of LCB are highly correlated with the temperature variation of the steam explosion pretreatment. Hemicellulose has been reported to begin hydrolysis at 150 °C (Garrote et al., 1999). Steam explosion can facilitate the hydrolysis of acetyl groups in hemicellulose components and the production of simpler sugars (Menardo et al., 2012). The lignin content can change greatly when the temperature of steam explosion pretreatment exceeds 160 °C. Several studies have shown that the lignin of lignocellulose is reduced after steam explosion pretreatment, while others have observed the opposite result (Bauer et al., 2014; Martin-Sampedro et al., 2011). Researchers have attributed the increase in lignin to pseudo-lignin formation, which cannot be distinguished from lignin by traditional measurement methods (Vivekanand et al., 2013). The increased intensity of steam explosion is beneficial to the degradation of lignocellulosic residues and the bioconversion efficiency of biomass without a significant loss of available substrates. However, although harsh pretreatment conditions can enlarge the reactive area, an excessive temperature and residence time may inhibit the bioconversion of LCB due to the formation of inhibitors (Lizasoain et al., 2017).

4.4. Ammonia fiber expansion

Ammonia fiber expansion (AFEX) is a novel pretreatment process, during which anhydrous liquid or gaseous ammonia is used to pretreat LCB in a pressurized container (Zhao et al., 2020). AFEX integrates chemical and physical processes, which is suitable for lignocellulose

Table 4
Physico-chemical pretreatment methods for the enhancement of biogas/methane production from LCB.

Physico-chemical pretreatment	Pretreatment condition	Biomass	AD condition	Maximum biogas/methane production under specific pretreatment conditions	Ref.
Liquid hot water (LHW) pretreatment	Pretreatment temperature: 160, 180, 200 °C; Pretreatment time: 20, 40, 60 min; Liquid-to-substrate ratio: 20:1, 25:1, 30:1;	Hybrid *Pennisetum*	Batch, 37 °C, 35 d	32.9 % increase in biomethane yield with LHW, 175 °C, 35 min	(Kang et al., 2020)
	Pretreatment temperature: 150–225 °C; Pretreatment time: 5–60 min; Liquid-to-substrate ratio: 30:1;	Wheat straw	Batch, 36 ± 2 °C, 42 d	62.9 % increase in methane yield with LHW, 175 °C, 30 min	(Shang et al., 2019)
	Pretreatment temperature: 140 °C; Pretreatment time: 100 s;	Yard waste	Batch, mesophilic temperature, 26 or 45 d	10.8 % increase in methane yield	(Panigrahi et al., 2019)
Wet oxidation	Pretreatment temperature: 185, 220 °C; Pretreatment time: 10, 15 min; Oxygen pressure: 0, 3, 12 bar; Na_2CO_3 addition: 0, 2 g/L;	Yard waste, food waste	Batch, 55 °C, 28 d	98.6 % increase in methane yield from pretreated yard waste with 220 °C, 15 min, 3 bar, 2 g/L	(Lissens et al., 2004)
	H_2O_2 concentration: 4.5 %, 7.5 %; Pretreatment temperature: 100, 110, 120 °C; Pretreatment time: 15, 30 min; Presoaking: 2 % NaOH, 4% NH_3;	Soybean straw	Batch, 37 °C, 30 d	62 % increase in methane yield from the co-digestion of pretreated straw and farm wastewater with 2 % wet NaOH presoaking, 4.5 % H_2O_2, 100 °C, 30 min	(Xiong et al., 2020)
	Oxygen loading: 0 %, 3 %, 6 %, 9 % (in the form of H_2O_2); Pretreatment temperature: 180 °C; Pretreatment time: 45 min;	Oil palm empty fruit bunches	Batch, 37 °C or 55 °C, 51 d	43 % increase in methane yield with 6 % oxygen at mesophilic condition	(Lee et al., 2020)
Steam explosion	Pretreatment temperature: 140–220 °C; Pretreatment time: 2, 5, 10, 15 min;	Corn stover	Batch, 37.5 °C, 49 d	22 % increase in methane yield with team explosion, 160 °C, 2 min	(Lizasoain et al., 2017)
	Pretreatment temperature: 162–240 °C; Pretreatment mass: 60, 78, 150 g; Pretreatment time: 12, 16, 30, 32 min; Pretreatment pressure: 6.5–29 bar; Severity parameter (S_0): 3.05, 3.54, 4.10, 4.32, 5.29 min;	Rice straw	Batch, mesophilic temperature, 35 d	32 % increase in methane yield with steam explosion, 206 °C, 150 g, 30 min, 18.5 bar (S_0 = 4.10)	(Steinbach et al., 2019)
	Pretreatment temperature: 210 °C; Pretreatment time: 10 min;	Wheat straw	Batch, 37 °C, 60 d	Approximately 20 % in methane yield with stem-exploded straw	(Sapci et al., 2013)
	Pretreatment temperature: 200, 210, 220 °C; Pretreatment time: 60, 90, 120, 240 s;	Rice straw	Batch, 38 °C, 21 d	51 % increase in biogas yield with steam explosion, 200 °C, 120 s	(Zhou et al., 2016)
	Pretreatment pressure: 1.2, 1.5, 1.8 MPa; Pretreatment time: 3, 6, 9, 12 min;	Rice straw	Batch, mesophilic temperature, 53 d	62.7 % increase in methane yield with steam explosion, 1.2 MPa, 3 min	(Siddhu et al., 2019)
Ammonia fiber expansion (AFEX)	Ammonia to dry matter ration: 2:1; Pretreatment time: 30 min; Pretreatment temperature: 102 °C; Pretreatment pressure: 325 psi;	Corn stover	Semi-continuous, 35 °C, HRT 20 d, digestion duration 75 d	22 % increase in biogas production from the co-digestion of AFEX-pretreated stover and dairy manure	(Rojas-Sossa et al., 2019)
	Non-uniform temperature range of 120–80 °C Pretreatment time: 60 min;	Sugarcane bagasse and cane leaf matter	Batch, pilot-scale, 37 °C, 55 d	26 % and 8 % increase in methane yield from cane leaf matter and and sugarcane bagasse	(Mokomele et al., 2019)

(*continued on next page*)

Table 4 (*continued*)

Physico-chemical pretreatment	Pretreatment condition	Biomass	AD condition	Maximum biogas/methane production under specific pretreatment conditions	Ref.
Supercritical carbon dioxide explosion (SC-CO_2)	Ammonia to biomass loading: 0.7 g NH_3/g DM Pretreatment temperature: 80, 90, 100 °C; Pretreatment time: 15, 30, 60 min; Pretreatment pressure: 5, 8, 10 MPa;	Napier grass	Batch, 35 °C, 24 d	156.5 % increase in methane yield with SC-CO_2, 100 °C, 60 min, 10 MPa	(Hude and Yadav, 2014)

pretreatment. Ammonia enhances cellulose decrystallization, hemicellulose depolymerization, and acetyl groups deacetylation under high temperatures and pressures. The inhibitors produced during AFEX pretreatment are less toxic to enzymatic hydrolysis and fermentation, and no further water-washing of AFEX-treated materials is required, thereby simplifying the operation procedure (Bals et al., 2014). In one study, AFEX pretreatment promoted the biodegradation of sugarcane residues, and compared with untreated materials, the methane yield of AFEX-pretreated sugarcane bagasse and the cane leaf matter increased by 8 % and 26 %, respectively (Mokomele et al., 2019). AFEX pretreatment can provide exogenous nitrogen to the fermentation system and consequently balance the C/N ratio of nitrogen-deficient substrates. The main function of AFEX pretreatment is to modify the cell wall structure rather than to remove hemicellulose and lignin (Chundawat et al., 2011). H_2O_2 is usually introduced into AFEX pretreatment to compensate for the low removal efficiency of hemicellulose and lignin *via* AFEX (Zhao et al., 2017b; Zhao et al., 2016). The sugar yield of H_2O_2/AFEX-pretreated corn stalk has been found to increase 3.31-fold as compared to untreated stalk (Zhao et al., 2016). To date, the mechanism of the H_2O_2/AFEX pretreatment of lignocellulosic residues remains unclear, and future studies could fruitfully explore this issue.

4.5. Supercritical carbon dioxide explosion

SC-CO_2 fluid exists at the temperature and pressure beyond the critical point of CO_2, in which case there is no difference between the gas and liquid phases (Capolupo and Faraco, 2016). The instantaneous release of CO_2 pressure can disrupt the cellulose structure, thus reducing cellulose crystallinity and increasing sugar production after enzymolysis (Dharmaraja et al., 2023). SC-CO_2 pretreatment is regarded as a promising method for LCB decomposition because it is clean, low-cost, and has higher diffusivity, among other benefits (Zhao et al., 2019). No harmful chemicals are produced during the pretreatment process, and CO_2 solvent can be easily separated from the product *via* the decompression process. The methane yield of SC-CO_2-pretreated Napier grass (100 °C, 60 min, 10 MPa) has been found to be increased by 156.5 % as compared to that of raw grass (Hude and Yadav, 2014). The moisture content of lignocellulosic materials has a significant influence on the SC-CO_2 pretreatment performance. One study revealed a remarkable increase in the sugar yields of SC-CO_2-pretreated materials with the increase of the lignocellulose moisture (Kim and Hong, 2001). In addition, the combination of SC-CO_2 pretreatment with other pretreatment methods provides a promising strategy for LCB utilization. The co-treatment method of SC-CO_2 and ultrasound, for example, has been reported to yield a 280 % increase in fermentable sugar from sugarcane bagasse as compared to the raw material (Benazzi et al., 2013). Fermentable sugar has been the primary value-added product in most studies on SC-CO_2 pretreatment of lignocellulosic residues, and considerable attention should be paid to the recovery of other energy products.

5. Biological pretreatment

There is increasing interest in biological pretreatment due to its merits of low toxicity and economic input as compared to physical and chemical pretreatments. Biological pretreatment can effectively disintegrate refractory lignocellulose into simple organic substances, shorten the subsequent fermentation period, and enhance the digestibility of feedstock. Methods based on bacterial consortium, fungal consortium, and enzymes are the main biological pretreatment methods for LCB (Table 5).

5.1. Microbial consortium-based pretreatment

A single bacterium or a consortium is used during this process to pretreat LCB, which has the merits of low energy input and the reduced formation of inhibitors. However, the conspicuous defect of microbial pretreatment is the long pretreatment period resulting from the long incubation time needed for sufficient microbial growth. The consortium used for LCB pretreatment is normally screened from the natural environment, for which decayed lignocellulosic wastes are often used as substrates. Microbial consortium pretreatment based on a thermophilic microbial consortium (MC1) has been reported to enhance biogas yield from cotton straw, specifically, MC1-pretreated material with a substrate load of 4 % (m/v) exhibited the maximum biogas yield after fermentation for 8 days (Yuan et al., 2016). Similarly, the same research group reported that the biogas yield of MC1-pretreated maize straw increased by 74.7 % compared with raw material, indicating that the sterilization of substrates before MC1 pretreatment was unnecessary (Hua et al., 2016).

Fungal consortium-based pretreatment is the other microbial pretreatment method used to improve the AD performance based on lignocellulosic residues (Rouches et al., 2016). In general, fungi used for lignocellulose degradation can be divided into brown-, white-, and soft-rot fungi (Wagner et al., 2018). Brown-rot fungi (BRF) can attack cellulose and hemicellulose and have a slight modification effect on lignin. White-rot fungi (WRF) and soft-rot fungi, particularly WRF, are mainly responsible for delignification. The unique enzymatic system in WRF attacks the phenolic structures in lignin and converts it to CO_2 (Sánchez, 2009). The fungal pretreatment performance of LCB is highly related to the species of fungi. Fungal pretreatment with three ligninolytic fungi (TF-1, TF-2, TF-3) screened from decayed wood has been found to shorten the lag phase time and increase the biogas productivity, and the biogas yield of TF-1-pretreated wheat straw was 5-fold higher than that of raw straw (Shah and Ullah, 2019). In another investigation, the biogas production of wheat straw pretreated with *Polyporus brumalis* BRFM985 achieved a 52 % increase as compared to the untreated material (Rouches et al., 2018). WRF pretreatment of corn silage with *Trametes versicolor* has proven to enhance the biogas productivity in an AcoD system (Tišma et al., 2018).

5.2. Enzyme-based pretreatment

The hydrolysis of cellulose and hemicellulose is pivotal for lignocellulose fermentation. The addition of exogenous hydrolase or oxidase can promote the decomposition of lignocellulose under anaerobic conditions (Cheng and Whang, 2022). Enzymes used for pretreatment can be classified into cellulose/hemicellulose-degrading enzymes and lignases on the basis of their activity on different components of

Table 5
Biological pretreatment methods for the enhancement of biogas/methane production from LCB.

Biological pretreatment	Pretreatment condition	Biomass	AD condition	Maximum biogas/methane production under specific pretreatment conditions	Ref.
Microbial pretreatment	A thermophilic microbial consortium (MC1); Pretreatment time: 2–10 d; Substrate concentration: 1 %–4 % (m/v);	Cotton stalk	Batch, 35 °C, 30 d	136 % increase in biogas yield with MC1, 8 d, 4 % (m/v)	(Yuan et al., 2016)
	A thermophilic microbial consortium (MC1); Pretreatment time: 5 or 10 d;	Rotted silage maize straw	Batch, 35 °C, 15 d	74.7 % increase in biogas yields with MC1, 5 d	(Hua et al., 2016)
	Microbial agents [0.01 % (w/w)]; Pretreatment time: 5, 10, 15, 20 d;	Corn straw	Batch, mesophilic temperature, 30 d	75.57 % increase in methane yield with 15 d	(Zhong et al., 2011)
	A microaerobic barley straw-adapted microbial (BSAM); Pretreatment time: 40 d; Loading ratio: BSAM to co-substrate was 1:7;	Barley straw and hay co-substrate	Batch, 33–36 °C, 35 d	40-fold increase in methane yield	(Raut et al., 2021)
	Fungus: *Pleurotus ostreatus* (DSM 11191); Pretreatment temperature: 28 °C; Pretreatment time: 10, 20, 30 d;	Rice straw	Batch, 37 °C, 30 d	67.6 % increase in methane yield with DSM, 30 d	(Mustafa et al., 2017)
	Lignolytic fungus: *Pleurotus ostreatus*; Pretreatment temperature: 28 °C; Pretreatment time: 21 d;	Wheat straw (WS) Pearl millet straw (PMS)	Batch, 37 °C, 60 d	38 % and 35 % increase in methane yield from WS and PMS, respectively	(Yadav et al., 2019)
	Lab-screened ligninolytic fungi: TF-1, TF-2, TF-3; Pretreatment temperature: 28 °C; Pretreatment time: 7 d;	Wheat straw	Batch, 50 ± 2 °C, 42 d	5-fold increase in biogas yield with TF-1	(Shah and Ullah, 2019)
Enzyme pretreatment	Enzyme: cellulase; Enzyme loads:10, 20, 30, 40 U/g TS; Pretreatment time: 8 d;	Rice Straw	Batch, 38 °C, 30 d	185.8 % increase in biogas yield with cellulase, 40 U/g TS	(Dai et al., 2018)
	Enzyme: cellulase generated by *Aspergillus niger*; Enzyme loads: 0, 10, 20, 30, 40, 50 FPU/g; Pretreatment time: 0, 12, 24, 36, 48, 60 h; Pretreatment solid content: 20, 30, 40, 50, 60, 70 g/L; Pretreatment temperature: 50 °C;	Corn stover	Batch, 37 °C, 18 d	36.9 % increase in biogas yield with cellulase, 30 FPU/g, 24 h, 60 g/L	(Wang et al., 2018)
	Enzyme: laccase (LA, 2 U/g), peroxidases (MnP, 5 U/g), versatile peroxidase (VP, 1.5 U/g), combination (COMB) of the 3 enzymes; Pretreatment temperature: 30 °C; Pretreatment time: 0, 6, 12, 24 h;	Corn stover	Batch, mesophilic temperature, 30 d	25 % increase in methane yield with LA, 24 h	(Schroyen et al., 2014)
	Enzyme: commercial enzymatic preparation Ultraflo® L, *Aspergillus* enzymatic extract; Enzyme loads: 0.2 U of feruloyl esterase per gram of dry substrate; Pretreatment temperature: 40 °C; Pretreatment time: 3 h;	Corn cob	Batch, mesophilic temperature, 32 d	14 % increase in methane yield with Ultraflo® L	(Pérez-Rodríguez et al., 2017a)

lignocellulose (Abraham et al., 2020). As previously reported, the cellulose content of rice straw decreased by 71.5 % at the end of cellulase pretreatment, and the biogas yield of 40 U/g-TS cellulose-pretreated straw was 2.9 times greater than that of raw straw (Dai et al., 2018). The cellulase pretreatment of corn stover with 30 FPU/g enzyme load, 24 h incubation time, and 60 g/L solid content contributed to a 36.9 % increase in biogas yield as compared to the raw material (Wang et al., 2018). Lignases can hydrolyze lignin into simpler monomers, thus making cellulose and hemicellulose more accessible to microorganisms. No inhibitors are generated during the enzyme pretreatment process with laccase, versatile peroxidase, and manganese peroxidase. The methane production of stover pretreated with laccase for 24 h and peroxidases for 6 h has been found to increase by 25 % and 17 %, respectively (Schroyen et al., 2014). Although the enzyme pretreatment of lignocellulosic wastes is characterized by the strengths of a short reaction time and relatively low sugar loss during pretreatment, the high economic cost and the environmental sensitivity of enzymes limit the practical application of enzyme pretreatment to a large extent.

Enzyme immobilization technologies, including surface binding, cross-linking, and entrapment, have been studied to enhance the catalytic activity, operational stability, and recovery and reusability of enzymes (Gao et al., 2022). Cross-linking involves covalent interactions between enzyme molecules instead of between the enzyme and the support carrier, which is known as carrier-free immobilization. Multiple support matrixes have been developed for enzyme immobilization based on surface binding and entrapment, such as glass (Eskandarloo and Abbaspourrad, 2018), biochar (Mo et al., 2020), alginate acid (Herizi et al., 2020), activated carbon (Santos et al., 2022), carbon nanotubes (Feng and Ji, 2011), etc. Nanomaterials have attracted public attention due to their peculiar crystallographic structure and surface properties facilitating a high enzyme loading capacity. Immobilization of enzyme-based biocatalysts, including cellulase, laccase, xylanase, etc., on supporting nanomaterials can effectively intensify the catalytic propertied and reusability of natural enzymes (Rai et al., 2019). The existing studies on enzyme-based biocatalysts mostly focus on recovering fermentable sugars from LCB, rather than biogas production. For instance, magnetic nanoparticle-immobilized cellulase has been certified to have higher enzyme activity and stability, and further benefit the bioconversion of rice straw (Kaur et al., 2021). In addition, nanogels and mesoporous silica nanospheres have been newly developed as support materials for protein immobilization (Gao et al., 2022; Kothalawala et al., 2022). To inspire more innovations, the development of more universal, low-cost,

and efficient enzyme agents and proper immobilization materials is warranted in the future.

6. New development on emerging pretreatment strategies for lignocellulosic biomass

In recent years, increasing attention has been given to emerging pretreatment methods with environmental-friendly and techno-economic feasibility concepts. Strategies of new-developed PEF, EBI, freezing-thawing, microaerobic, and nanotechnical pretreatment are discussed herein for further improving the bioconversion efficiency of LCB to methane and reducing the use of chemicals and the generation of by-products, thereby providing beneficial inspiration and guidance for future research (Table 6).

6.1. Pulsed electric field

PEF is demonstrated to be a non-thermal and energy-efficient method in comparison with conventional thermal methods (Basak et al., 2023). By report, PEF can not only enhance cell permeability and promote the release of organics but also improve the decomposition of complex organic matters and benefit subsequent utilization (Kovačić et al., 2021). PEF pretreatment has been shown to be effective in the destruction of lignocellulosic structure and thereby improving subsequent fermentation efficiency (Paudel et al., 2017). For instance, PEF pretreatment yielded a 16 % increase in methane yield compared with the untreated maize silage with the electric field intensity of 20 kV/cm and frequency of 10 kHz for 180 s (Szwarc and Szwarc, 2020). The methane production of PEF-pretreated rapeseed straw was 14 % higher than that of raw straw with electric pulse of 40 kV, pulse width of 50 μs, and frequency of 5 kHz for 5 min (Szwarc and Głowacka, 2021). Research on the impact of PEF on biogas production is still in its infancy and focuses on the treatment of wastewater and sewage sludge (Kovačić et al., 2021). It is necessary to strengthen our focus on the PEF pretreatment of LCB for biogas production both in a laboratory scale and large-scale. Moreover, the economic effectiveness and environmental impact of PEF pretreatment on biogas production in real AD plants is of great importance for its future industrial exploitation.

6.2. Electron beam irradiation

EBI pretreatment achieved by a linear accelerator can directly disrupt the complex structure of lignocellulose, and the produced free radicals will further contribute to the chain scission, decrystallization, and depolymerization of LCB (Olatunji et al., 2021). Several works have confirmed the positive effect of EBI pretreatment on fermentable sugar production from LCB (Hassan et al., 2018; Yang et al., 2015b). EBI pretreatment of birch could degrade lignin, benefit cellulose

Table 6
New development on emerging pretreatment strategies for lignocellulosic biomass.

Physical pretreatment	Pretreatment condition	LCB	AD/Enzymatic hydrolysis	Results	Ref.
Pulsed electric field (PEF)	Rectangular electrical pulses; Voltage of 40 kV amplitude; Width of 50 μs; Frequency of 5 kHz	Chicken manure	Batch, 37 °C, 24 d	19.4 % increase in methane yield	(Szwarc et al., 2023)
	Pulse generator of 2.5 kV and 0.24 A; $W_{spec.} = 153$ k/kg; $t_{PEF} = 1$ s; $N = 14$ trains	Grape pomace	Batch, 37 °C, 30 d	4 % increase in methane yield	(El Achkar et al., 2018)
	Rectangular electrical pulses; Voltage of −40 kV; Pulse duration of 50 μs; Frequency of 10 kHz; Retention time of 180 s	Maize Silage	Batch, 37 °C, 24 d	14 % increase in biogas production	(Szwarc and Szwarc, 2020)
	Rectangular electrical pulses; Voltage of 40 kV; Pulse duration of 50 μs; Frequency of 5 kHz; Retention time of 5 min	Rapeseed straw	Batch, 35 °C, 24 d	14 % increase in methane production	(Szwarc and Głowacka, 2021)
Electron beam irradiation (EBI)	Electron beam energy of 10 MeV; Irradiation doses rate of 50 kGy/pass or 20 kGy/pass; Irradiation doses of 1200 kGy of birch	Birch	Cellulase activity of 30 U/mg, 40 °C, 48 h	256.5 % increase in total sugar production of birch	(Fei et al., 2020)
	Total dose exposure of 500 kGy; Electron beams with an energy of 1 MeV and 7.4 mA.	Korean *Miscanthus sinensis*	A commercial enzyme mixture, 50 °C, 48 h	26 % increase in glucose conversion	(Yang et al., 2015b)
Freezing-thawing pretreatment	Freezing-thawing pretreatment for 21d under naturally frozen outdoors; Solid-to-liquid ratio of 1:6	Corn stalk	Batch, 35 °C, 55 d	40.5 % increase in biomethane yield	(Li et al., 2019)
	Combined freezing-thawing (−20 °C, 5 d) and ammonia (2 wt%), with 70 % moisture content	Corn stover	Batch, 35 °C, 55 d	41.1 % increase in biomethane yield	(Yuan et al., 2019)
	Combined freezing-thawing (−20 °C/20 °C, 7 d) and KOH (4 %/TS)	Corn straw	Batch, 35 °C, 50 d	39.8 % increase in biomethane yield	(Yang et al., 2023c)
Microaerobic pretreatment	Oxygen loading of 5 mL/g VS substrate with microbial agent of biogas slurry at 55 °C	Corn straw	Batch, 37 °C, 60 d	16.2 % increase in methane yield	(Fu et al., 2015)
	Oxygen loading of 15 mL/g VS with microbial agent of digestate effluent at 37 °C for 4.5 d	Paper waste	Batch, 37 °C, 55–60 d	10.1 % increase in methane yield	(Song et al., 2021)
	Oxygen loading of 5 mL/g VS substrate with microbial agent of biogas slurry at 55 °C	*Miscanthus floridulus*	Batch, 35 °C, 45–50 d	25.5 % increase in methane yield	(Fu et al., 2018)
	Dissolved oxygen concentration of 0.1–1 mg/L with microbial agent of fresh cow manure. 12 h pretreatment of substrate at pH 8	Corn stover	Batch, 38 °C, 20 d	16.6 % increase in methane yield	(Xu et al., 2021)
Nanomaterials-based pretreatment	Alkylsulfonic and butylcarboxylic acid functionalized magnetic nanoparticles (Fe_3O_4-MNPs-Si-AS and Fe_3O_4-MNPs-Si-BCOOH) with the concentration of 100, 200, 300, 400, 500 mg/g straw at 120 °C and 15 psi pressure for 15 min	Sugarcane straw	N/A	Xylose production increased in concentration dependent; 71.1-fold and 64.2-fold than raw straw	(Ingle et al., 2020a)
	Fe_3O_4-MNPs@Si@AS and Fe_3O_4-MNPs@Si@BCOOH with the concentration of 100, 200, 300, 400, 500 mg/g substrate at 120 °C and 15 psi pressure for 15 min	Sugarcane bagasse	N/A	Xylose production increased in concentration dependent; 67.3-fold and 66.7-fold than raw straw	(Ingle et al., 2020b)
	Cerium doped iron oxide nanoparticles possessed unique enzyme mimicking characteristics; In combination with H_2O_2 at 25 °C and 200 rpm for 24 h with solid loading of 20 % (w/v)	Corn cob	N/A	Maximum delignification of 44 %	(Rajak et al., 2021)

decrystallization and depolymerization, and disrupt cellulose molecules, thus enhancing substrate enzymatic hydrolysis and sugar production (Fei et al., 2020). However, EBI pretreatment is mainly useful in cellulose depolymerization, and it is often used in combination with other pretreatment methods (Olatunji et al., 2021). For instance, the production of reducing sugar from LCB was obviously promoted by alkali and acid soaking assisted EBI pretreatment (Karthika et al., 2013; Xiang et al., 2017). So far, few studies have investigated the effect of EBI pretreatment or EBI-involved co-treatment on LCB biogas production, which should be a priority for the future research.

6.3. Freezing-thawing pretreatment

Freezing-thawing pretreatment involves mixing cellulosic biomass with water, freezing it below 0 °C for a period of time, and then thawing it to room temperature, and the entire process will be repeated a few times (Guan et al., 2022). The moisture in LCB can increase as a result of freezing, and the structure cellulose-hemicellulose-lignin matrix will be opened, and therefore increase the surface area of cellulose. Freezing-thawing pretreatment has been proved to improve the biogas productivity of corn stalk, and after pretreatment for 21 days, the methane yield increased by 40.5 % over the raw material with solid-to-liquid ratio of 1:6 (Li et al., 2019). This pretreatment can be performed in natural conditions such as winter without any additional energy input. Current research often uses freezing-thawing in combination with alkaline reagents, e.g., KOH and ammonia, to further improve the bioconversion efficiency of straw waste. The co-treatment of freezing-thawing (−20 °C, 5 days) and ammonia (2 wt%) yield a 41.08 % increase in methane yield from corn stover in comparison with the untreated group (Yuan et al., 2019). Furthermore, combined freezing-thawing (−20 °C/20 °C, 7 days) and KOH (4 %/TS) has been found to achieve a 39.8 % increase in methane production compared with raw corn straw, and manifest greater energy sustainability and economics (Yang et al., 2023c). Future work should put more stress on the optimization of co-treatment method, and find more suitable methods for LCB co-treatment with freezing-thawing.

6.4. Microaerobic pretreatment

Interestingly, the presence of trace oxygen in AD systems is reported to enhance feedstock hydrolysis, system stability, and hydrogen-sulfide stripping (Fu et al., 2023). Two common methods for this purpose include microaerobic pretreatment and micro-aeration during AD process. Although both methods can improve AD performance, microaerobic pretreatment is easier to operate than micro-aeration (Cai et al., 2021). Accordingly, microaerobic pretreatment of LCB is regarded as a promising strategy to intensify its further AD process. The positive effect of microaerobic pretreatment on LCB anaerobic fermentation has been confirmed. A previous study found that the methane yield increased by 16.24 % from thermophilic microaerobic pretreated of corn straw with oxygen load of 5 mL/g VS (Fu et al., 2015). The main function of microaerobic pretreatment is to create an appropriate living condition for hydrolytic bacteria, and thus to intensify the LCB hydrolysis. For example, microaerobic pretreatment was shown to promote the hydrolysis as well as acidification of corn stover by enriching functional bacteria and fungi (Xu et al., 2021). Microaerobic pretreatment has drawn extensive attention in terms of its economic and environmental benefits, during which merely trace amount of oxygen and/or air is introduced into the treatment system with simple operation. Microaerobic pretreatment of *Miscanthus floridulus* was proved to be economically viable over alkali H_2O_2, NaOH, HCl, and hot-water pretreatments, with the methane yield increased by 25.5 % compared to the untreated material (Fu et al., 2018). Despite recent advances, the study on LCB microaerobic pretreatment is still in its infancy, the effect of microaerobic pretreatment on LCB AD performance varies depending on oxygen supply mode and concentration, pretreatment time, etc. More work needs to be done on the oxygen-dosing optimization and the response mechanism of microbial communities to oxygen and the information of metabolic regulation.

6.5. Nanomaterials-based pretreatment

Interestingly, artificial enzyme mimics are reported to be a promising substitute for natural enzyme, which overcome the shortcomings of natural enzyme such as low stability, high cost and complex preparation process (Jiang et al., 2019). Specifically, nanocatalysts are a collection of nanomaterials with multivalent catalytic sites and an average particle size of <100 nm, which have wider environmental suitability and longer life cycle than their natural analogues (Gao et al., 2007). Nanocatalysts possess the properties of both chemical catalysts as well as biocatalysts. A previous study has shown that MOF-808, one type of enzyme mimicking nanoparticles, could hydrolyze macromolecular organics during sludge AD process (Liu et al., 2022). In addition, the Ce-MOF-based nanocatalysts with the ligand of fumaric acid was also reported to cleave phosphoester bonds, amide bonds, and glycosidic bonds, and even hydrolyze biofilms composed of polysaccharides, proteins, nucleic acids, and lipids (Li et al., 2022). Nanocatalysts with high catalytic performance and stability have been developed rapidly in promoting LCB hydrolysis in recent years. Nanomaterials-based pretreatment has been demonstrated to benefit LCB disintegration and the production of biogas, biohydrogen, fermentable sugar, etc. (Srivastava et al., 2023). Notably, the utilization of nanomaterials can reduce the process cost as the high recoverability and reusability of these reagents. Nanocatalyst of shear hybrid alkaline is often used for LCB pretreatment, which can enter the lignocellulosic cell wall and react with biomass components to produce carbohydrates (Lay et al., 2022). Besides, acid-functionalized magnetic nanoparticles are proved to enhance LCB hydrolysis and biofuels production, and the strong magnetic feature of such nanoparticles makes the recycling and reuse of materials more feasible (Chandel et al., 2022). For instance, the presence of acid-functionalized magnetic nanoparticles (Fe_3O_4-Si-AS and Fe_3O_4-Si-BCOOH) favored xylose production from sugarcane straw and bagasse compared with H_2SO_4-pretreaed and untreated materials (Ingle et al., 2020a; Ingle et al., 2020b). The development of cerium-doped iron oxide nanoparticles was certified effective for lignin depolymerization in corn cob (Rajak et al., 2021). The rapid development of nanomaterials has promoted the upgrading of LCB pretreatment methods, and nanomaterials-based pretreatment is increasingly perceived as an effective alternative option to traditional pretreatment methods due to its high techno-economic feasibility and ecological friendliness.

7. Challenges and prospects

Various pretreatment methods have been developed for biomethane production from LCB. It is important for the selection of pretreatment methods to assess the economic effectiveness and environmental impact of different pretreatment methods. To date, many laboratory-scale experiments based on straw pretreatment have been performed for evaluating the economic and environmental benefits of various pretreatment methods. For instance, 3 % KOH was demonstrated to be the best choice for switchgrass based on the comprehensive analysis of economic, energy and environmental benefits over NaOH, $Ca(OH)_2$, H_2O_2, HCl, H_2SO_4, and steam explosion (Shen et al., 2020). Comparing with acid, alkali, and LHW pretreatments, the steam explosion was the optimal method to pretreat Icelandic lignocellulose for its following AD, considering the economic, energy, as well as environmental impacts (Safarian and Unnthorsson, 2018). Combined pretreatment methods can mitigate the disadvantages of single pretreatment and achieve higher pretreatment efficiency. Physical-chemical pretreatment is reported to be a more flexible and well-studied method for methane production from straw wastes, and although the high-cost of pure chemicals reduces its practical application, the utilization of chemical reagents from wastes

can greatly decline pretreatment cost and environmental pollution (Ma et al., 2022). Moreover, wet oxidation/explosion integrates the advantages of physical and chemical pretreatments with the merits of eco-friendly and low-cost and has gained increasing notice.

TEA is a novel approach of evaluating in economic consequences of different process models and engineering and technical designs (Fig. 3) (Ahmad et al., 2018). TEA refers to a comprehensive model that can calculate, compare, and demonstrate the profitability of a technological process in order to achieve the best combination of techno-economy. Despite several studies have been performed to determine the profitability of AD based on methane production from agriculture wastes, TEA reports of entire processes including pretreatment and subsequent AD remain scarce (Ahmad et al., 2018; Vasco-Correa et al., 2018). LCA is a vital tool for analyzing environmental evaluation of various pretreatment methods. Thermo-chemo-sonic pretreatment of yard waste was reported to be an eco-friendly method based on a gate-to-gate LCA study, which negatively impacted on carbon emissions and carbon footprint (Panigrahi et al., 2022). In fact, TEA coupled with LCA are increasingly being regarded as necessary tools for comparing the environmental implications, economic effectiveness, and sustainability of various pretreatments and fermentation processes, and a great deal more effort is recommended to conduct this field. However, most studies are conducted under specific conditions, and it is difficult to make a comprehensive comparison of pretreatment conditions, feedstock properties, and economic input and output under different experimental frameworks. As experimental data accumulates, artificial intelligence technologies, such as machine learning, can illustrate multivariate relationships between dependent and independent variables *via* the construction of data-driven predictive models, thus providing an effective means for the selection of pretreatment methods (Fig. 3) (Zhu et al., 2023). In addition, the selection of pretreatment method needs to take into account the operability as well as compatibility of following AD process. Straw is generally comminuted before use and plenty of water is needed to wash the substrate, which will considerably increase the economic input and the loss of resources. The direct application of pretreated LCB for AD process is a more realistic solution, putting more demand on the development and application of pretreatment technology.

8. Conclusions

Lignocellulose pretreatment is crucially important in accessible surface area increase, lignin decomposition, and cellulose-hemicellulose hydrolysis. The effectiveness of pretreatment methods greatly influences the efficiency of lignocellulosic waste treatment. Physical pretreatment is highly effective in enhancing biomethanation potential, but the high cost limits its practical application. Chemical pretreatment has proven to enhance the decomposition of lignocellulose and alter the structural compositions. However, the byproducts produced during the pretreatment processes may cause secondary pollution, and the use of exogenous chemicals increases costs. In recent years, increasing attention has been given to biological pretreatment, which exhibits good degradation performance for LCB with lower costs and no secondary pollution. The construction of highly efficient synthetic consortia and the intrinsic limitations of natural enzymes hinder the practical application of biological pretreatment. The development of nanozymes and enzyme immobilization technique paves the way for exploring biological pretreatment of LCB. Furthermore, the combined pretreatment methods can fully exploit the advantages of each method, compensating for the deficiency of individual treatment, and have become a hot topic in current research due to its potential for large-scale application. The mechanisms of different pretreatment methods, the substrate types and

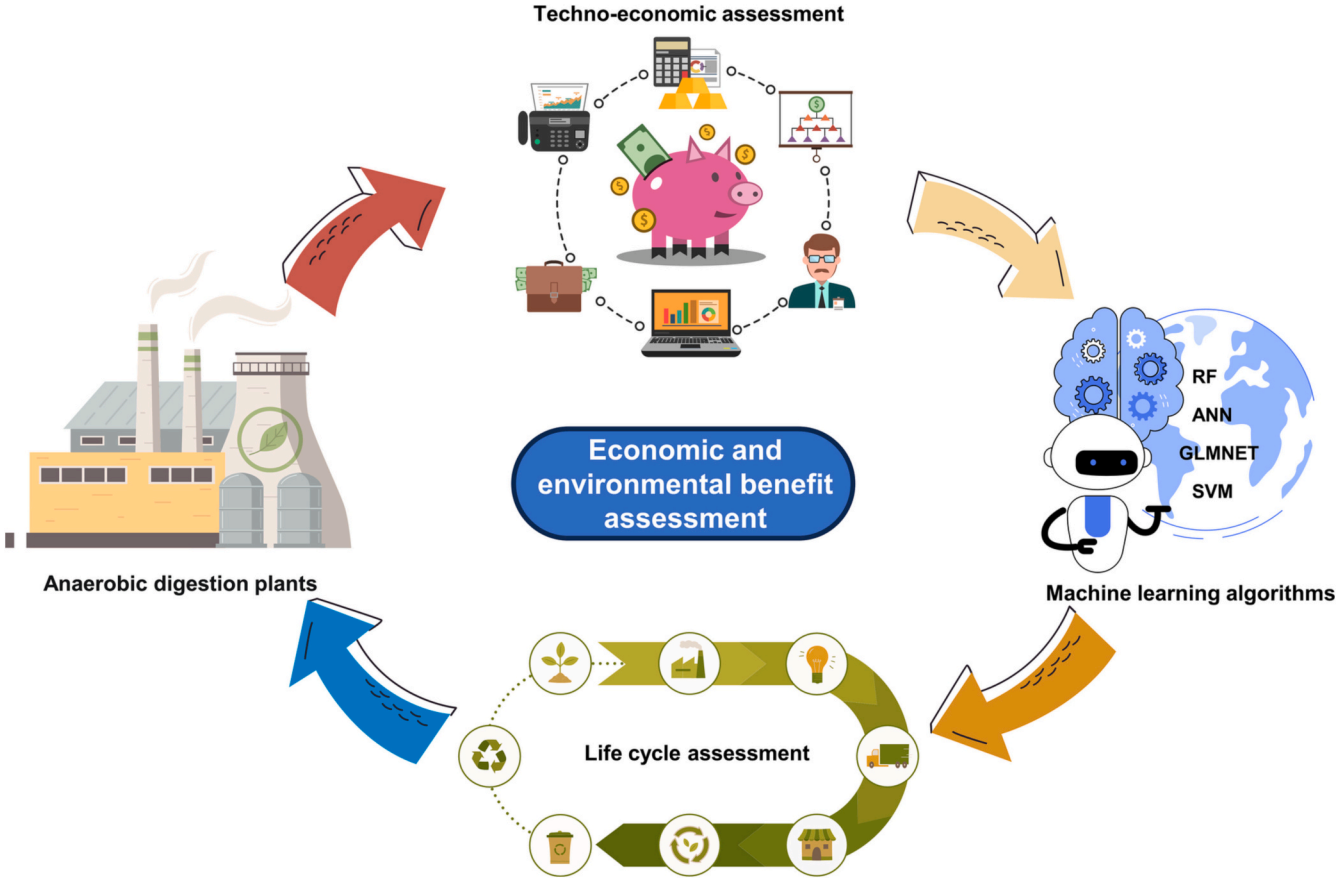


Fig. 3. Techno-economic and environmental benefits of anaerobic digestion based on techno-economic analysis, machine learning, and life cycle assessment.

compositions, the experimental conditions, should be comprehensively considered before selecting a pretreatment method. While pretreatment technology has made significant progress, the development of process equipment to make pretreatment more economical and efficient on a larger scale needs to catch up.

CRediT authorship contribution statement

Jing Wang: Investigation; Writing-Original Draft; Writing-Reviewing and Editing; Visualization;

Dongmei Ma: Investigation and Visualization;

Yu Lou: Investigation and Visualization;

Jun Ma: Formal analysis and Investigation;

Defeng Xing: Formal analysis; Writing-Original Draft; Writing-Reviewing and Editing; Project administration; Funding acquisition; Supervision;

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (No. U22A20443), the State Key Laboratory of Urban Water Resource and Environment (Harbin Institute of Technology) (No. 2022TS06), the Fundamental Research Funds for the Central Universities (HIT.OCEF.2022049), the Interdisciplinary Research Foundation of HIT, and the Heilongjiang Key Research and Development Program (2022ZX02C19).

References

Abraham, A., Mathew, A.K., Park, H., et al., 2020. Pretreatment strategies for enhanced biogas production from lignocellulosic biomass. Bioresour. Technol. 301, 122725. https://doi.org/10.1016/j.biortech.2019.122725.

Aguilar-Reynosa, A., Romani, A., Rodriguez-Jasso, R.M., et al., 2017. Comparison of microwave and conduction-convection heating autohydrolysis pretreatment for bioethanol production. Bioresour. Technol. 243, 273–283. https://doi.org/10.1016/j.biortech.2017.06.096.

Ahmad, F., Silva, E.L., Varesche, M.B.A., 2018. Hydrothermal processing of biomass for anaerobic digestion – a review. Renew. Sustain. Energy Rev. 98, 108–124. https://doi.org/10.1016/j.rser.2018.09.008.

Akobi, C., Yeo, H., Hafez, H., et al., 2016. Single-stage and two-stage anaerobic digestion of extruded lignocellulosic biomass. Appl. Energy 184, 548–559. https://doi.org/10.1016/j.apenergy.2016.10.039.

Alonso, D.M., Wettstein, S.G., Mellmer, M.A., et al., 2013. Integrated conversion of hemicellulose and cellulose from lignocellulosic biomass. Energy Environ. Sci. 6 (1), 76–80. https://doi.org/10.1039/c2ee23617f.

Alvira, P., Tomas-Pejo, E., Ballesteros, M., et al., 2010. Pretreatment technologies for an efficient bioethanol production process based on enzymatic hydrolysis: a review. Bioresour. Technol. 101 (13), 4851–4861. https://doi.org/10.1016/j.biortech.2009.11.093.

Amin, F.R., Khalid, H., Li, W., et al., 2021. Enhanced methane production and energy potential from rice straw by employing microaerobic pretreatment via anaerobic digestion. J. Clean. Prod. 296 https://doi.org/10.1016/j.jclepro.2021.126434.

Amiri, H., Karimi, K., Zilouei, H., 2014. Organosolv pretreatment of rice straw for efficient acetone, butanol, and ethanol production. Bioresour. Technol. 152, 450–456. https://doi.org/10.1016/j.biortech.2013.11.038.

Baeta, B.E.L., Cordeiro, P.H.M., Passos, F., et al., 2017. Steam explosion pretreatment improved the biomethanization of coffee husks. Bioresour. Technol. 245 (Pt A), 66–72. https://doi.org/10.1016/j.biortech.2017.08.110.

Bals, B.D., Gunawan, C., Moore, J., et al., 2014. Enzymatic hydrolysis of pelletized AFEX™-treated corn stover at high solid loadings. Biotechnol. Bioeng. 111 (2), 264–271. https://doi.org/10.1002/bit.25022.

Basak, B., Kumar, R., Bharadwaj, A.V.S.L.S., et al., 2023. Advances in physicochemical pretreatment strategies for lignocellulose biomass and their effectiveness in bioconversion for biofuel production. Bioresour. Technol. 369 https://doi.org/10.1016/j.biortech.2022.128413.

Bauer, A., Lizasoain, J., Theuretzbacher, F., et al., 2014. Steam explosion pretreatment for enhancing biogas production of late harvested hay. Bioresour. Technol. 166, 403–410. https://doi.org/10.1016/j.biortech.2014.05.025.

Benazzi, T., Calgaroto, S., Astolfi, V., et al., 2013. Pretreatment of sugarcane bagasse using supercritical carbon dioxide combined with ultrasound to improve the enzymatic hydrolysis. Enzyme Microb. Technol. 52 (4), 247–250. https://doi.org/10.1016/j.enzmictec.2013.02.001.

Bhatia, S.K., Jagtap, S.S., Bedekar, A.A., et al., 2020. Recent developments in pretreatment technologies on lignocellulosic biomass: effect of key parameters, technological improvements, and challenges. Bioresour. Technol. 300, 122724. https://doi.org/10.1016/j.biortech.2019.122724.

Bianco, F., Senol, H., Papirio, S., 2021. Enhanced lignocellulosic component removal and biomethane potential from chestnut shell by a combined hydrothermal-alkaline pretreatment. Sci. Total Environ. 762, 144178. https://doi.org/10.1016/j.scitotenv.2020.144178.

Bjerre, A., Olesen, A., Fernqvist, T., et al., 1996. Pretreatment of wheat straw using combined wet oxidation and alkaline hydrolysis resulting in convertible cellulose and hemicellulose. Biotechnol. Bioeng. 49 (5), 568–577. https://doi.org/10.1002/(SICI)1097-0290(19960305)49:5<568::AID-BIT10>3.0.CO;2-6.

Cai, Y., Zheng, Z., Schäfer, F., et al., 2021. A review about pretreatment of lignocellulosic biomass in anaerobic digestion: achievement and challenge in Germany and China. J. Clean. Prod. 299 https://doi.org/10.1016/j.jclepro.2021.126885.

Capolupo, L., Faraco, V., 2016. Green methods of lignocellulose pretreatment for biorefinery development. Appl. Microbiol. Biotechnol. 100 (22), 9451–9467. https://doi.org/10.1007/s00253-016-7884-y.

Chandel, H., Kumar, P., Chandel, A.K., et al., 2022. Biotechnological advances in biomass pretreatment for bio-renewable production through nanotechnological intervention. Biomass Convers. Biorefin. https://doi.org/10.1007/s13399-022-02746-0.

Chang, K.L., Chen, X.M., Han, Y.J., et al., 2016. Synergistic effects of surfactant-assisted ionic liquid pretreatment rice straw. Bioresour. Technol. 214, 371–375. https://doi.org/10.1016/j.biortech.2016.04.113.

Chen, Q.L., Ding, J., Li, C.Y., et al., 2020. Microbial functional attributes, rather than taxonomic attributes, drive top soil respiration, nitrification and denitrification processes. Sci. Total Environ. 734, 139479. https://doi.org/10.1016/j.scitotenv.2020.139479.

Chen, W.-H., Nižetić, S., Sirohi, R., et al., 2022. Liquid hot water as sustainable biomass pretreatment technique for bioenergy production: a review. Bioresour. Technol. 344, 126207. https://doi.org/10.1016/j.biortech.2021.126207.

Chen, X., Zhang, Y., Gu, Y., et al., 2014. Enhancing methane production from rice straw by extrusion pretreatment. Appl. Energy 122, 34–41. https://doi.org/10.1016/j.apenergy.2014.01.076.

Cheng, H.-H., Whang, L.-M., 2022. Resource recovery from lignocellulosic wastes via biological technologies: advancements and prospects. Bioresour. Technol. 343, 126097. https://doi.org/10.1016/j.biortech.2021.126097.

Chufo, A., Yuan, H.R., Zou, D.X., et al., 2015. Biomethane production and physicochemical characterization of anaerobically digested teff (*Eragrostis tef*) straw pretreated by sodium hydroxide. Bioresour. Technol. 181, 214–219. https://doi.org/10.1016/j.biortech.2015.01.054.

Chundawat, S.P.S., Donohoe, B.S., da Costa Sousa, L., et al., 2011. Multi-scale visualization and characterization of lignocellulosic plant cell wall deconstruction during thermochemical pretreatment. Energy Environ. Sci. 4 (3), 973–984. https://doi.org/10.1039/c0ee00574f.

Croce, S., Wei, Q., D'Imporzano, G., et al., 2016. Anaerobic digestion of straw and corn stover: the effect of biological process optimization and pre-treatment on total biomethane yield and energy performance. Biotechnol. Adv. 34 (8), 1289–1304. https://doi.org/10.1016/j.biotechadv.2016.09.004.

Dai, B.-L., Guo, X.-J., Yuan, D.-H., et al., 2018. Comparison of different pretreatments of rice straw substrate to improve biogas production. Waste Biomass Valorization 9 (9), 1503–1512. https://doi.org/10.1007/s12649-017-9950-9.

Dai, X., Hua, Y., Dai, L., et al., 2019. Particle size reduction of rice straw enhances methane production under anaerobic digestion. Bioresour. Technol. 293, 122043. https://doi.org/10.1016/j.biortech.2019.122043.

Deivayanai, V.C., Yaashikaa, P.R., Senthil Kumar, P., et al., 2022. A comprehensive review on the biological conversion of lignocellulosic biomass into hydrogen: pretreatment strategy, technology advances and perspectives. Bioresour. Technol. 365 https://doi.org/10.1016/j.biortech.2022.128166.

Dharmaraja, J., Shobana, S., Arvindnarayan, S., et al., 2023. Lignocellulosic biomass conversion via greener pretreatment methods towards biorefinery applications. Bioresour. Technol. 369, 128328. https://doi.org/10.1016/j.biortech.2022.128328.

Domanski, J., Marchut-Mikolajczyk, O., Cieciura-Wloch, W., et al., 2020. Production of methane, hydrogen and ethanol from *Secale cereale* L. straw pretreated with sulfuric acid. Molecules 25 (4), 1013. https://doi.org/10.3390/molecules25041013.

Dong, L., Cao, G., Zhao, L., et al., 2018. Alkali/urea pretreatment of rice straw at low temperature for enhanced biological hydrogen production. Bioresour. Technol. 267, 71–76. https://doi.org/10.1016/j.biortech.2018.05.055.

Dong, P., Li, L., Zhen, F., et al., 2017. Comparison of dry and wet milling pre-treatment methods for improving the anaerobic digestion performance of the *Pennisetum* hybrid. RSC Adv. 7 (21), 12610–12619. https://doi.org/10.1039/c6ra27822a.

Du, J., Qian, Y., Xi, Y., et al., 2019. Hydrothermal and alkaline thermal pretreatment at mild temperature in solid state for physicochemical properties and biogas production from anaerobic digestion of rice straw. Renew. Energy 139, 261–267. https://doi.org/10.1016/j.renene.2019.01.097.

El Achkar, J.H., Lendormi, T., Salameh, D., et al., 2018. Influence of pretreatment conditions on lignocellulosic fractions and methane production from grape pomace. Bioresour. Technol. 247, 881–889. https://doi.org/10.1016/j.biortech.2017.09.182.
Elalami, D., Carrere, H., Abdelouahdi, K., et al., 2020. Mild microwaves, ultrasonic and alkaline pretreatments for improving methane production: impact on biochemical and structural properties of olive pomace. Bioresour. Technol. 299, 122591. https://doi.org/10.1016/j.biortech.2019.122591.
Eskandarloo, H., Abbaspourrad, A., 2018. Production of galacto-oligosaccharides from whey permeate using beta-galactosidase immobilized on functionalized glass beads. Food Chem. 251, 115–124. https://doi.org/10.1016/j.foodchem.2018.01.068.
Fang, Y., Si, B., Qiu, J., et al., 2020. Bioconversion of bamboo shoot shell to methane assisted by microwave irradiation and fungus metabolism. Sci. Total Environ. 724, 138268. https://doi.org/10.1016/j.scitotenv.2020.138268.
Fei, X., Jia, W., Wang, J., et al., 2020. Study on enzymatic hydrolysis efficiency and physicochemical properties of cellulose and lignocellulose after pretreatment with electron beam irradiation. Int. J. Biol. Macromol. 145, 733–739. https://doi.org/10.1016/j.ijbiomac.2019.12.232.
Feng, W., Ji, P., 2011. Enzymes immobilized on carbon nanotubes. Biotechnol. Adv. 29 (6), 889–895. https://doi.org/10.1016/j.biotechadv.2011.07.007.
Fu, S., Lian, S., Angelidaki, I., et al., 2023. Micro-aeration: an attractive strategy to facilitate anaerobic digestion. Trends Biotechnol. 41 (5), 714–726. https://doi.org/10.1016/j.tibtech.2022.09.008.
Fu, S.-F., Wang, F., Yuan, X.-Z., et al., 2015. The thermophilic (55°C) microaerobic pretreatment of corn straw for anaerobic digestion. Bioresour. Technol. 175, 203–208. https://doi.org/10.1016/j.biortech.2014.10.072.
Fu, S.-F., Chen, K.-Q., Zhu, R., et al., 2018. Improved anaerobic digestion performance of *Miscanthus floridulus* by different pretreatment methods and preliminary economic analysis. Energy Convers. Manag. 159, 121–128. https://doi.org/10.1016/j.enconman.2018.01.014.
Gao, J., Chen, L., Yuan, K., et al., 2013. Ionic liquid pretreatment to enhance the anaerobic digestion of lignocellulosic biomass. Bioresour. Technol. 150, 352–358. https://doi.org/10.1016/j.biortech.2013.10.026.
Gao, L., Zhuang, J., Nie, L., et al., 2007. Intrinsic peroxidase-like activity of ferromagnetic nanoparticles. Nat. Nanotechnol. 2 (9), 577–583. https://doi.org/10.1038/nnano.2007.260.
Gao, Y., Shah, K., Kwok, I., et al., 2022. Immobilized fungal enzymes: innovations and potential applications in biodegradation and biosynthesis. Biotechnol. Adv. 57, 107936. https://doi.org/10.1016/j.biotechadv.2022.107936.
Garrote, G., Dominguez, H., Parajo, J.C., 1999. Hydrothermal processing of lignocellulosic materials. Holz Roh Werkst. 57, 191–202.
Gould, J.M., 1984. Alkaline peroxide delignification of agricultural residues to enhance enzymatic saccharification. Biotechnol. Bioeng. 26 (1), 46–52. https://doi.org/10.1002/bit.260260110.
Gu, Y., Zhang, Y., Zhou, X., 2015. Effect of $Ca(OH)_2$ pretreatment on extruded rice straw anaerobic digestion. Bioresour. Technol. 196, 116–122. https://doi.org/10.1016/j.biortech.2015.07.004.
Guan, R., Yuan, H., Yuan, S., et al., 2022. Current development and perspectives of anaerobic bioconversion of crop stalks to biogas: a review. Bioresour. Technol. 349 https://doi.org/10.1016/j.biortech.2021.126615.
Guo, M., Song, W., Buhain, J., 2015. Bioenergy and biofuels: history, status, and perspective. Renew. Sustain. Energy Rev. 42, 712–725. https://doi.org/10.1016/j.rser.2014.10.013.
Hassan, S.S., Williams, G.A., Jaiswal, A.K., 2018. Emerging technologies for the pretreatment of lignocellulosic biomass. Bioresour. Technol. 262, 310–318. https://doi.org/10.1016/j.biortech.2018.04.099.
He, Z.-W., Liu, W.-Z., Gao, Q., et al., 2018. Potassium ferrate addition as an alternative pre-treatment to enhance short-chain fatty acids production from waste activated sludge. Bioresour. Technol. 247, 174–181. https://doi.org/10.1016/j.biortech.2017.09.073.
Herizi, A., Rachid, S., Djaffar, D., et al., 2020. Optimization and immobilization of alpha-amylase from *Bacillus subtilis* in calcium alginate and calcium alginate–cellulosic residue beads. Microbiol. Res. 11 (1), 8458. https://doi.org/10.4081/mr.2020.8458.
Herrmann, C., Heiermann, M., Idler, C., et al., 2012. Particle size reduction during harvesting of crop feedstock for biogas production I: effects on ensiling process and methane yields. Bioenergy Res. 5 (4), 926–936. https://doi.org/10.1007/s12155-012-9206-2.
Hesami, S.M., Zilouei, H., Karimi, K., et al., 2015. Enhanced biogas production from sunflower stalks using hydrothermal and organosolv pretreatment. Ind. Crops Prod. 76, 449–455. https://doi.org/10.1016/j.indcrop.2015.07.018.
Hjorth, M., Granitz, K., Adamsen, A.P., et al., 2011. Extrusion as a pretreatment to increase biogas production. Bioresour. Technol. 102 (8), 4989–4994. https://doi.org/10.1016/j.biortech.2010.11.128.
Hoang, A.T., Nguyen, X.P., Duong, X.Q., et al., 2023. Steam explosion as sustainable biomass pretreatment technique for biofuel production: characteristics and challenges. Bioresour. Technol. 385, 129398. https://doi.org/10.1016/j.biortech.2023.129398.
Hu, J., Guo, B., Li, Z., et al., 2020. Revealing the mechanisms for potassium ferrate affecting methane production from anaerobic digestion of waste activated sludge. Bioresour. Technol. 317, 124022. https://doi.org/10.1016/j.biortech.2020.124022.
Hua, B., Dai, J., Liu, B., et al., 2016. Pretreatment of non-sterile, rotted silage maize straw by the microbial community MC1 increases biogas production. Bioresour. Technol. 216, 699–705. https://doi.org/10.1016/j.biortech.2016.06.001.
Hude, M.P., Yadav, G.D., 2014. Process intensification in methane generation during anaerobic digestion of Napier grass using supercritical carbon dioxide combined with acid hydrolysis pre-treatment. Can. J. Chem. Eng. 92 (12), 2176–2184. https://doi.org/10.1002/cjce.22080.
Ingle, A.P., Philippini, R.R., de Souza Melo, Y.C., et al., 2020a. Acid-functionalized magnetic nanocatalysts mediated pretreatment of sugarcane straw: an eco-friendly and cost-effective approach. Cellulose 27 (12), 7067–7078. https://doi.org/10.1007/s10570-020-03262-y.
Ingle, A.P., Philippini, R.R., Silvério da Silva, S., 2020b. Pretreatment of sugarcane bagasse using two different acid-functionalized magnetic nanoparticles: a novel approach for high sugar recovery. Renew. Energy 150, 957–964. https://doi.org/10.1016/j.renene.2019.11.146.
Izumi, K., Okishio, Y.-k., Nagao, N., et al., 2010. Effects of particle size on anaerobic digestion of food waste. Int. Biodeterior. Biodegr. 64 (7), 601–608. https://doi.org/10.1016/j.ibiod.2010.06.013.
Jackowiak, D., Bassard, D., Pauss, A., et al., 2011. Optimisation of a microwave pretreatment of wheat straw for methane production. Bioresour. Technol. 102 (12), 6750–6756. https://doi.org/10.1016/j.biortech.2011.03.107.
Jaffar, M., Pang, Y., Yuan, H., et al., 2016. Wheat straw pretreatment with KOH for enhancing biomethane production and fertilizer value in anaerobic digestion. Chin. J. Chem. Eng. 24 (3), 404–409. https://doi.org/10.1016/j.cjche.2015.11.005.
Jiang, D., Ni, D., Rosenkrans, Z.T., et al., 2019. Nanozyme: new horizons for responsive biomedical applications. Chem. Soc. Rev. 48 (14), 3683–3704. https://doi.org/10.1039/c8cs00718g.
Jing, Q., LÜ, X., 2007. Kinetics of non-catalyzed decomposition of D-xylose in high temperature liquid water. Chin. J. Chem. Eng. 15 (5), 666–669. https://doi.org/10.1016/s1004-9541(07)60143-8.
Kabir, M.M., Niklasson, C., Taherzadeh, M.J., et al., 2014. Biogas production from lignocelluloses by N-methylmorpholine-N-oxide (NMMO) pretreatment: effects of recovery and reuse of NMMO. Bioresour. Technol. 161, 446–450. https://doi.org/10.1016/j.biortech.2014.03.107.
Kang, X., Zhang, Y., Lin, R., et al., 2020. Optimization of liquid hot water pretreatment on hybrid *Pennisetum* anaerobic digestion and its effect on energy efficiency. Energy Conv. Manag. 210, 112718. https://doi.org/10.1016/j.enconman.2020.112718.
Karimi, M., Jenkins, B., Stroeve, P., 2014. Ultrasound irradiation in the production of ethanol from biomass. Renew. Sustain. Energy Rev. 40, 400–421. https://doi.org/10.1016/j.rser.2014.07.151.
Karthika, K., Arun, A.B., Melo, J.S., et al., 2013. Hydrolysis of acid and alkali presoaked lignocellulosic biomass exposed to electron beam irradiation. Bioresour. Technol. 129, 646–649. https://doi.org/10.1016/j.biortech.2012.12.048.
Karunanithy, C., Muthukumarappan, K., 2011. Optimization of switchgrass and extruder parameters for enzymatic hydrolysis using response surface methodology. Ind. Crops Prod. 33 (1), 188–199. https://doi.org/10.1016/j.indcrop.2010.10.008.
Kaur, P., Taggar, M.S., Kalia, A., 2021. Characterization of magnetic nanoparticle–immobilized cellulases for enzymatic saccharification of rice straw. Biomass Convers. Biorefin. 11 (3), 955–969. https://doi.org/10.1007/s13399-020-00628-x.
Khor, W.C., Rabaey, K., Vervaeren, H., 2015. Low temperature calcium hydroxide treatment enhances anaerobic methane production from (extruded) biomass. Bioresour. Technol. 176, 181–188. https://doi.org/10.1016/j.biortech.2014.11.037.
Kim, I., Lee, B., Park, J.Y., et al., 2014. Effect of nitric acid on pretreatment and fermentation for enhancing ethanol production of rice straw. Carbohydr. Polym. 99, 563–567. https://doi.org/10.1016/j.carbpol.2013.08.092.
Kim, K.H., Hong, J., 2001. Supercritical CO_2 pretreatment of lignocellulose enhances enzymatic cellulose hydrolysis. Bioresour. Technol. 77 (2), 139–144. https://doi.org/10.1016/S0960-8524(00)00147-4.
Kim, Y., Mosier, N.S., Ladisch, M.R., 2009. Enzymatic digestion of liquid hot water pretreated hybrid poplar. Biotechnol. Prog. 25 (2), 340–348. https://doi.org/10.1002/btpr.137.
Korai, R.M., Wachemo, A.C., Yue, L., et al., 2020. Effect of ultrasonic application during KOH pretreatment and anaerobic process on digestion performance of wheat straw. RSC Adv. 10 (16), 9290–9298. https://doi.org/10.1039/d0ra00525h.
Kothalawala, S.G., Jiao, J., Speight, R., et al., 2022. Pore architecture influences the enzyme immobilization performance of mesoporous silica nanospheres. Microporous Mesoporous Mater. 338, 111963. https://doi.org/10.1016/j.micromeso.2022.111963.
Kovačić, Đ., Rupčić, S., Kralik, D., et al., 2021. Pulsed electric field: an emerging pretreatment technology in a biogas production. Waste Manag. 120, 467–483. https://doi.org/10.1016/j.wasman.2020.10.009.
Kumar, A.K., Sharma, S., 2017. Recent updates on different methods of pretreatment of lignocellulosic feedstocks: a review. Bioresour. Bioprocess. 4 (1), 7. https://doi.org/10.1186/s40643-017-0137-9.
Lara-Serrano, M., Sáez Angulo, F., Negro, M.J., et al., 2018. Second-generation bioethanol production combining simultaneous fermentation and saccharification of IL-pretreated barley straw. ACS Sustain. Chem. Eng. 6 (5), 7086–7095. https://doi.org/10.1021/acssuschemeng.8b00953.
Lay, C.H., Dharmaraja, J., Shobana, S., et al., 2022. Lignocellulose biohydrogen towards net zero emission: a review on recent developments. Bioresour. Technol. 364, 128084. https://doi.org/10.1016/j.biortech.2022.128084.
Lee, J.T.E., Khan, M.U., Tian, H., et al., 2020. Improving methane yield of oil palm empty fruit bunches by wet oxidation pretreatment: mesophilic and thermophilic anaerobic digestion conditions and the associated global warming potential effects. Energy Conv. Manag. 225, 113438. https://doi.org/10.1016/j.enconman.2020.113438.
Lee, J.W., Jeffries, T.W., 2011. Efficiencies of acid catalysts in the hydrolysis of lignocellulosic biomass over a range of combined severity factors. Bioresour. Technol. 102 (10), 5884–5890. https://doi.org/10.1016/j.biortech.2011.02.048.
Li, C., Knierim, B., Manisseri, C., et al., 2010. Comparison of dilute acid and ionic liquid pretreatment of switchgrass: biomass recalcitrance, delignification and enzymatic saccharification. Bioresour. Technol. 101 (13), 4900–4906. https://doi.org/10.1016/j.biortech.2009.10.066.

Li, D., Liu, S., Mi, L., et al., 2015a. Effects of feedstock ratio and organic loading rate on the anaerobic mesophilic co-digestion of rice straw and pig manure. Bioresour. Technol. 187, 120–127. https://doi.org/10.1016/j.biortech.2015.03.040.
Li, J., Wachemo, A.C., Yuan, H., et al., 2019. Natural freezing-thawing pretreatment of corn stalk for enhancing anaerobic digestion performance. Bioresour. Technol. 288, 121518. https://doi.org/10.1016/j.biortech.2019.121518.
Li, J., Wachemo, A.C., Yu, G.Q., et al., 2020. Enhanced anaerobic digestion performance of corn stalk pretreated with freezing-thawing and ammonia: an experimental and theoretical study. J. Clean. Prod. 247, 12. https://doi.org/10.1016/j.jclepro.2019.119112.
Li, L., Chen, C., Zhang, R., et al., 2015b. Pretreatment of corn stover for methane production with the combination of potassium hydroxide and calcium hydroxide. Energy Fuels 29 (9), 5841–5846. https://doi.org/10.1021/acs.energyfuels.5b01170.
Li, L., He, J., Xin, X., et al., 2018. Enhanced bioproduction of short-chain fatty acids from waste activated sludge by potassium ferrate pretreatment. Chem. Eng. J. 332, 456–463. https://doi.org/10.1016/j.cej.2017.09.103.
Li, S., Zhou, Z., Tie, Z., et al., 2022. Data-informed discovery of hydrolytic nanozymes. Nat. Commun. 13 (1), 827. https://doi.org/10.1038/s41467-022-28344-2.
Lissens, G., Thomsen, A.B., De Baere, L., et al., 2004. Thermal wet oxidation improves anaerobic biodegradability of raw and digested biowaste. Environ. Sci. Technol. 38 (12), 3418–3424. https://doi.org/10.1021/es035092h.
Liu, H., Xu, Y., Geng, H., et al., 2022. Contributions of MOF-808 to methane production from anaerobic digestion of waste activated sludge. Water Res. 220, 118653. https://doi.org/10.1016/j.watres.2022.118653.
Liu, X., Zicari, S.M., Liu, G., et al., 2015. Improving the bioenergy production from wheat straw with alkaline pretreatment. Biosyst. Eng. 140, 59–66. https://doi.org/10.1016/j.biosystemseng.2015.09.006.
Liu, X., Huang, X., Wu, Y., et al., 2020a. Activation of nitrite by freezing process for anaerobic digestion enhancement of waste activated sludge: performance and mechanisms. Chem. Eng. J. 387, 124147. https://doi.org/10.1016/j.cej.2020.124147.
Liu, Z., Li, L., Liu, C., et al., 2018. Pretreatment of corn straw using the alkaline solution of ionic liquids. Bioresour. Technol. 260, 417–420. https://doi.org/10.1016/j.biortech.2018.03.117.
Liu, Z., Zhou, A., Liu, H., et al., 2020b. Extracellular polymeric substance decomposition linked to hydrogen recovery from waste activated sludge: role of peracetic acid and free nitrous acid co-pretreatment in a prefermentation-bioelectrolysis cascading system. Water Res. 176, 115724. https://doi.org/10.1016/j.watres.2020.115724.
Lizasoain, J., Trulea, A., Gittinger, J., et al., 2017. Corn stover for biogas production: effect of steam explosion pretreatment on the gas yields and on the biodegradation kinetics of the primary structural compounds. Bioresour. Technol. 244 (Pt 1), 949–956. https://doi.org/10.1016/j.biortech.2017.08.042.
Ma, S., Li, Y., Li, J., et al., 2022. Features of single and combined technologies for lignocellulose pretreatment to enhance biomethane production. Renew. Sustain. Energy Rev. 165. https://doi.org/10.1016/j.rser.2022.112606.
Mancini, G., Papirio, S., Lens, P.N.L., et al., 2018a. Anaerobic digestion of lignocellulosic materials using ethanol-organosolv pretreatment. Environ. Eng. Sci. 35 (9), 953–960. https://doi.org/10.1089/ees.2018.0042.
Mancini, G., Papirio, S., Lens, P.N.L., et al., 2018b. Increased biogas production from wheat straw by chemical pretreatments. Renew. Energy 119, 608–614. https://doi.org/10.1016/j.renene.2017.12.045.
Martinez-Jimenez, F.D., Macie Pinto, M.P., Mudhoo, A., et al., 2017. Influence of ultrasound irradiation pre-treatment in biohythane generation from the thermophilic anaerobic co-digestion of sugar production residues. J. Environ. Chem. Eng. 5 (4), 3749–3758. https://doi.org/10.1016/j.jece.2017.07.030.
Martin-Sampedro, R., Capanema, E.A., Hoeger, I., et al., 2011. Lignin changes after steam explosion and laccase-mediator treatment of eucalyptus wood chips. J. Agric. Food Chem. 59 (16), 8761–8769. https://doi.org/10.1021/jf201605f.
Menardo, S., Bauer, A., Theuretzbacher, F., et al., 2012. Biogas production from steam-exploded miscanthus and utilization of biogas energy and CO_2 in greenhouses. Bioenergy Res. 6 (2), 620–630. https://doi.org/10.1007/s12155-012-9280-5.
Mikulski, D., Klosowski, G., 2018. Efficiency of dilute sulfuric acid pretreatment of distillery stillage in the production of cellulosic ethanol. Bioresour. Technol. 268, 424–433. https://doi.org/10.1016/j.biortech.2018.08.005.
Millati, R., Wikandari, R., Ariyanto, T., et al., 2020. Pretreatment technologies for anaerobic digestion of lignocelluloses and toxic feedstocks. Bioresour. Technol. 304, 122998. https://doi.org/10.1016/j.biortech.2020.122998.
Mo, H., Qiu, J., Yang, C., et al., 2020. Preparation and characterization of magnetic polyporous biochar for cellulase immobilization by physical adsorption. Cellulose 27 (9), 4963–4973. https://doi.org/10.1007/s10570-020-03125-6.
Mokomele, T., da Costa Sousa, L., Balan, V., et al., 2019. Incorporating anaerobic co-digestion of steam exploded or ammonia fiber expansion pretreated sugarcane residues with manure into a sugarcane-based bioenergy-livestock nexus. Bioresour. Technol. 272, 326–336. https://doi.org/10.1016/j.biortech.2018.10.049.
Monlau, F., Barakat, A., Steyer, J.P., et al., 2012. Comparison of seven types of thermo-chemical pretreatments on the structural features and anaerobic digestion of sunflower stalks. Bioresour. Technol. 120, 241–247. https://doi.org/10.1016/j.biortech.2012.06.040.
Mota, T.R., Oliveira, D.M., Morais, G.R., et al., 2019. Hydrogen peroxide-acetic acid pretreatment increases the saccharification and enzyme adsorption on lignocellulose. Ind. Crops Prod. 140, 111657. https://doi.org/10.1016/j.indcrop.2019.111657.
Motasemi, F., Ani, F.N., 2012. A review on microwave-assisted production of biodiesel. Renew. Sustain. Energy Rev. 16 (7), 4719–4733. https://doi.org/10.1016/j.rser.2012.03.069.
Mozhiarasi, V., Weichgrebe, D., Srinivasan, S.V., 2020. Enhancement of methane production from vegetable, fruit and flower market wastes using extrusion as pretreatment and kinetic modeling. Water Air Soil Pollut. 231 (3), 126. https://doi.org/10.1007/s11270-020-04469-2.
Mustafa, A.M., Poulsen, T.G., Xia, Y., et al., 2017. Combinations of fungal and milling pretreatments for enhancing rice straw biogas production during solid-state anaerobic digestion. Bioresour. Technol. 224, 174–182. https://doi.org/10.1016/j.biortech.2016.11.028.
Mustafa, A.M., Li, H., Radwan, A.A., et al., 2018. Effect of hydrothermal and $Ca(OH)_2$ pretreatments on anaerobic digestion of sugarcane bagasse for biogas production. Bioresour. Technol. 259, 54–60. https://doi.org/10.1016/j.biortech.2018.03.028.
Nargotra, P., Sharma, V., Gupta, M., et al., 2018. Application of ionic liquid and alkali pretreatment for enhancing saccharification of sunflower stalk biomass for potential biofuel-ethanol production. Bioresour. Technol. 267, 560–568. https://doi.org/10.1016/j.biortech.2018.07.070.
Neshat, S.A., Mohammadi, M., Najafpour, G.D., et al., 2017. Anaerobic co-digestion of animal manures and lignocellulosic residues as a potent approach for sustainable biogas production. Renew. Sustain. Energy Rev. 79, 308–322. https://doi.org/10.1016/j.rser.2017.05.137.
Oladi, S., Aita, G.M., 2017. Optimization of liquid ammonia pretreatment variables for maximum enzymatic hydrolysis yield of energy cane bagasse. Ind. Crops Prod. 103, 122–132. https://doi.org/10.1016/j.indcrop.2017.02.023.
Olatunji, K.O., Ahmed, N.A., Ogunkunle, O., 2021. Optimization of biogas yield from lignocellulosic materials with different pretreatment methods: a review. Biotechnol. Biofuels 14 (1). https://doi.org/10.1186/s13068-021-02012-x.
Olugbemide, A.D., Lajide, L., Adebayo, A., et al., 2019. Enhanced biogas production from rice husk through solid-state chemical pretreatments. Waste Biomass Valorization 11 (6), 2397–2407. https://doi.org/10.1007/s12649-018-00567-9.
Panepinto, D., Genon, G., 2016. Analysis of the extrusion as a pretreatment for the anaerobic digestion process. Ind. Crops Prod. 83, 206–212. https://doi.org/10.1016/j.indcrop.2015.12.044.
Panigrahi, S., Sharma, H.B., Dubey, B.K., 2019. Overcoming yard waste recalcitrance through four different liquid hot water pretreatment techniques – structural evolution, biogas production and energy balance. Biomass Bioenergy 127, 105268. https://doi.org/10.1016/j.biombioe.2019.105268.
Panigrahi, S., Tiwari, B.R., Brar, S.K., et al., 2022. Thermo-chemo-sonic pretreatment of lignocellulosic waste: evaluating anaerobic biodegradability and environmental impacts. Bioresour. Technol. 361, 127675. https://doi.org/10.1016/j.biortech.2022.127675.
Paudel, S.R., Banjara, S.P., Choi, O.K., et al., 2017. Pretreatment of agricultural biomass for anaerobic digestion: current state and challenges. Bioresour. Technol. 245 (Pt A), 1194–1205. https://doi.org/10.1016/j.biortech.2017.08.182.
Perez-Rodriguez, N., Garcia-Bernet, D., Dominguez, J.M., 2016. Effects of enzymatic hydrolysis and ultrasounds pretreatments on corn cob and vine trimming shoots for biogas production. Bioresour. Technol. 221, 130–138. https://doi.org/10.1016/j.biortech.2016.09.013.
Pérez-Rodríguez, N., García-Bernet, D., Domínguez, J.M., 2017a. Extrusion and enzymatic hydrolysis as pretreatments on corn cob for biogas production. Renew. Energy 107, 597–603. https://doi.org/10.1016/j.renene.2017.02.030.
Pérez-Rodríguez, N., García-Bernet, D., Domínguez, J.M., 2017b. Faster methane production after sequential extrusion and enzymatic hydrolysis of vine trimming shoots. Environ. Chem. Lett. 16 (1), 295–299. https://doi.org/10.1007/s10311-017-0668-5.
Periyasamy, S., Karthik, V., Senthil Kumar, P., et al., 2022. Chemical, physical and biological methods to convert lignocellulosic waste into value-added products: a review. Environ. Chem. Lett. 20 (2), 1129–1152. https://doi.org/10.1007/s10311-021-01374-w.
Petersson, A., Thomsen, M.H., Hauggaard-Nielsen, H., et al., 2007. Potential bioethanol and biogas production using lignocellulosic biomass from winter rye, oilseed rape and faba bean. Biomass Bioenergy 31 (11), 812–819. https://doi.org/10.1016/j.biombioe.2007.06.001.
Purwandari, F.A., Sanjaya, A.P., Millati, R., et al., 2013. Pretreatment of oil palm empty fruit bunch (OPEFB) by N-methylmorpholine-N-oxide (NMMO) for biogas production: structural changes and digestion improvement. Bioresour. Technol. 128, 461–466. https://doi.org/10.1016/j.biortech.2012.10.088.
Qiu, J., Tian, D., Shen, F., et al., 2018. Bioethanol production from wheat straw by phosphoric acid plus hydrogen peroxide (PHP) pretreatment via simultaneous saccharification and fermentation (SSF) at high solid loadings. Bioresour. Technol. 268, 355–362. https://doi.org/10.1016/j.biortech.2018.08.009.
Rabemanolontsoa, H., Saka, S., 2016. Various pretreatments of lignocellulosics. Bioresour. Technol. 199, 83–91. https://doi.org/10.1016/j.biortech.2015.08.029.
Radha, M., Murugesan, A.G., 2017. Enhanced dark fermentative biohydrogen production from marine macroalgae *Padina tetrastromatica* by different pretreatment processes. Biofuel Res. J. 4 (1), 551–558. https://doi.org/10.18331/brj2017.4.1.5.
Rai, M., Ingle, A.P., Pandit, R., et al., 2019. Emerging role of nanobiocatalysts in hydrolysis of lignocellulosic biomass leading to sustainable bioethanol production. Catal. Rev. Sci. Eng. 61 (1), 1–26. https://doi.org/10.1080/01614940.2018.1479503.
Rajak, R.C., Saha, P., Singhvi, M., et al., 2021. An eco-friendly biomass pretreatment strategy utilizing reusable enzyme mimicking nanoparticles for lignin depolymerization and biofuel production. Green Chem. 23 (15), 5584–5599. https://doi.org/10.1039/d1gc01456k.
Raut, M.P., Pandhal, J., Wright, P.C., 2021. Effective pretreatment of lignocellulosic co-substrates using barley straw-adapted microbial consortia to enhanced biomethanation by anaerobic digestion. Bioresour. Technol. 321, 124437. https://doi.org/10.1016/j.biortech.2020.124437.

Ravindran, R., Jaiswal, A.K., 2016. A comprehensive review on pre-treatment strategy for lignocellulosic food industry waste: challenges and opportunities. Bioresour. Technol. 199, 92–102. https://doi.org/10.1016/j.biortech.2015.07.106.
Rodrigues, R.C.L.B., Green Rodrigues, B., Vieira Canettieri, E., et al., 2022. Comprehensive approach of methods for microstructural analysis and analytical tools in lignocellulosic biomass assessment – a review. Bioresour. Technol. 348 https://doi.org/10.1016/j.biortech.2021.126627.
Rojas-Sossa, J.P., Zhong, Y., Valenti, F., et al., 2019. Effects of ammonia fiber expansion (AFEX) treated corn stover on anaerobic microbes and corresponding digestion performance. Biomass Bioenergy 127, 105263. https://doi.org/10.1016/j.biombioe.2019.105263.
Rouches, E., Herpoël-Gimbert, I., Steyer, J.P., et al., 2016. Improvement of anaerobic degradation by white-rot fungi pretreatment of lignocellulosic biomass: a review. Renew. Sustain. Energy Rev. 59, 179–198. https://doi.org/10.1016/j.rser.2015.12.317.
Rouches, E., Zhou, S., Sergent, M., et al., 2018. Influence of white-rot fungus *Polyporus brumalis* BRFM 985 culture conditions on the pretreatment efficiency for anaerobic digestion of wheat straw. Biomass Bioenergy 110, 75–79. https://doi.org/10.1016/j.biombioe.2018.01.018.
Sabeeh, M., Zeshan, Liaquat, R., et al., 2020. Effect of alkaline and alkaline-photocatalytic pretreatment on characteristics and biogas production of rice straw. Bioresour. Technol. 309, 123449. https://doi.org/10.1016/j.biortech.2020.123449.
Safarian, S., Unnthorsson, R., 2018. An assessment of the sustainability of lignocellulosic bioethanol production from wastes in Iceland. Energies 11 (6). https://doi.org/10.3390/en11061493.
Salapa, I., Katsimpouras, C., Topakas, E., et al., 2017. Organosolv pretreatment of wheat straw for efficient ethanol production using various solvents. Biomass Bioenergy 100, 10–16. https://doi.org/10.1016/j.biombioe.2017.03.011.
Sánchez, C., 2009. Lignocellulosic residues: biodegradation and bioconversion by fungi. Biotechnol. Adv. 27 (2), 185–194. https://doi.org/10.1016/j.biotechadv.2008.11.001.
Santos, M.P.F., Porfirio, M.C.P., Junior, E.C.S., et al., 2022. Pepsin immobilization: influence of carbon support functionalization. Int. J. Biol. Macromol. 203, 67–79. https://doi.org/10.1016/j.ijbiomac.2022.01.135.
Sapci, Z., 2013. The effect of microwave pretreatment on biogas production from agricultural straws. Bioresour. Technol. 128, 487–494. https://doi.org/10.1016/j.biortech.2012.09.094.
Sapci, Z., Morken, J., Linjordet, R., 2013. An investigation of the enhancement of biogas yields from lignocellulosic material using two pretreatment methods: microwave irradiation and steam explosion. BioResources 8 (2), 1976–1985. https://doi.org/10.15376/biores.8.2.1976-1985.
Sarto, S., Hildayati, R., Syaichurrozi, I., 2019. Effect of chemical pretreatment using sulfuric acid on biogas production from water hyacinth and kinetics. Renew. Energy 132, 335–350. https://doi.org/10.1016/j.renene.2018.07.121.
Schroyen, M., Vervaeren, H., Van Hulle, S.W.H., et al., 2014. Impact of enzymatic pretreatment on corn stover degradation and biogas production. Bioresour. Technol. 173, 59–66. https://doi.org/10.1016/j.biortech.2014.09.030.
Senol, H., 2020. Anaerobic digestion of hazelnut (*Corylus colurna*) husks after alkaline pretreatment and determination of new important points in logistic model curves. Bioresour. Technol. 300, 8. https://doi.org/10.1016/j.biortech.2019.122660.
Şenol, H., 2020. Identification of new critical points for logistics model in cumulative methane yield curves after co-digestion of apple pulp and chicken manure with sulphuric acid pretreatment and a new modelling study. Int. J. Energy Res. 44 (7), 6078–6087. https://doi.org/10.1002/er.5343.
Shah, T.A., Ullah, R., 2019. Pretreatment of wheat straw with ligninolytic fungi for increased biogas productivity. Int. J. Environ. Sci. Technol. 16 (11), 7497–7508. https://doi.org/10.1007/s13762-019-02277-8.
Shang, G., Zhang, C., Wang, F., et al., 2019. Liquid hot water pretreatment to enhance the anaerobic digestion of wheat straw-effects of temperature and retention time. Environ. Sci. Pollut. Res. 26 (28), 29424–29434. https://doi.org/10.1007/s11356-019-06111-z.
Shen, J., Zheng, Q., Zhang, R., et al., 2019. Co-pretreatment of wheat straw by potassium hydroxide and calcium hydroxide: methane production, economics, and energy potential analysis. J. Environ. Manag. 236, 720–726. https://doi.org/10.1016/j.jenvman.2019.01.046.
Shen, J., Zhang, J., Wang, W., et al., 2020. Assessment of pretreatment effects on anaerobic digestion of switchgrass: economics-energy-environment (3E) analysis. Ind. Crops Prod. 145 https://doi.org/10.1016/j.indcrop.2019.111957.
Shikinaka, K., Otsuka, Y., Nakamura, M., et al., 2018. Utilization of lignocellulosic biomass via novel sustainable process. J. Oleo Sci. 67 (9), 1059–1070. https://doi.org/10.5650/jos.ess18075.
Shrivastava, A., Sharma, R.K., 2022. Lignocellulosic biomass based microbial fuel cells: performance and applications. J. Clean. Prod. 361 https://doi.org/10.1016/j.jclepro.2022.132269.
Siddhu, M.A.H., Li, W., He, Y., et al., 2019. Steam explosion pretreatment of rice straw to improve structural carbohydrates anaerobic digestibility for biomethanation. Environ. Sci. Pollut. Res. Int. 26 (22), 22189–22196. https://doi.org/10.1007/s11356-019-05382-w.
Song, C., Li, W.W., Cai, F.F., et al., 2021. Anaerobic and microaerobic pretreatment for improving methane production from paper waste in anaerobic digestion. Front. Microbiol. 12 https://doi.org/10.3389/fmicb.2021.688290.
Song, Z., GaiheYang, Liu, X., et al., 2014. Comparison of seven chemical pretreatments of corn straw for improving methane yield by anaerobic digestion. PloS One 9 (4), e93801. https://doi.org/10.1371/journal.pone.0093801.
Song, Z.-L., Yag, G.-H., Feng, Y.-Z., et al., 2013. Pretreatment of rice straw by hydrogen peroxide for enhanced methane yield. J. Integr. Agric. 12 (7), 1258–1266. https://doi.org/10.1016/s2095-3119(13)60355-x.
Srivastava, N., Singh, R., Srivastava, M., et al., 2023. Impact of nanomaterials on sustainable pretreatment of lignocellulosic biomass for biofuels production: an advanced approach. Bioresour. Technol. 369, 128471. https://doi.org/10.1016/j.biortech.2022.128471.
Steinbach, D., Wust, D., Zielonka, S., et al., 2019. Steam explosion conditions highly influence the biogas yield of rice straw. Molecules 24 (19), 3492. https://doi.org/10.3390/molecules24193492.
Sun, C., Xie, Y., Hou, F., et al., 2020. Enhancement on methane production and anaerobic digestion stability via co-digestion of microwave-Ca$(OH)_2$ pretreated sugarcane rind slurry and kitchen waste. J. Clean. Prod. 264, 121731. https://doi.org/10.1016/j.jclepro.2020.121731.
Sun, H., Liao, C., Chen, L., et al., 2023. Potential for volatile fatty acid production via anaerobically-fermenting rice straw pretreated with silage effluent and phenyllactic acid. Bioresour. Technol. 369, 128355. https://doi.org/10.1016/j.biortech.2022.128355.
Sun, S., Sun, S., Cao, X., et al., 2016. The role of pretreatment in improving the enzymatic hydrolysis of lignocellulosic materials. Bioresour. Technol. 199, 49–58. https://doi.org/10.1016/j.biortech.2015.08.061.
Sun, Y., Cheng, J., 2002. Hydrolysis of lignocellulosic materials for ethanol production: a review. Bioresour. Technol. 83 (1), 1–11. https://doi.org/10.1016/S0960-8524(01)00212-7.
Syaichurrozi, I., Villta, P.K., Nabilah, N., et al., 2019. Effect of sulfuric acid pretreatment on biogas production from *Salvinia molesta*. J. Environ. Chem. Eng. 7 (1), 9. https://doi.org/10.1016/j.jece.2018.102857.
Szwarc, D., Głowacka, K., 2021. Increasing the biogas potential of rapeseed straw using pulsed electric field pre-treatment. Energies 14 (24). https://doi.org/10.3390/en14248307.
Szwarc, D., Szwarc, K., 2020. Use of a pulsed electric field to improve the biogas potential of maize silage. Energies 14 (1). https://doi.org/10.3390/en14010119.
Szwarc, D., Nowicka, A., Zieliński, M., 2023. Comparison of the effects of pulsed electric field disintegration and ultrasound treatment on the efficiency of biogas production from chicken manure. Appl. Sci. 13 (14) https://doi.org/10.3390/app13148154.
Tang, C., Shan, J., Chen, Y., et al., 2017. Organic amine catalytic organosolv pretreatment of corn stover for enzymatic saccharification and high-quality lignin. Bioresour. Technol. 232, 222–228. https://doi.org/10.1016/j.biortech.2017.02.041.
Teghammar, A., Karimi, K., Sárvári Horváth, I., et al., 2012. Enhanced biogas production from rice straw, triticale straw and softwood spruce by NMMO pretreatment. Biomass Bioenergy 36, 116–120. https://doi.org/10.1016/j.biombioe.2011.10.019.
Tian, Y.L., Zhang, H.Y., Mi, X.Y., et al., 2016. Research on anaerobic digestion of corn stover enhanced by dilute acid pretreatment: mechanism study and potential utilization in practical application. J. Renew. Sustain. Energy 8 (2), 14. https://doi.org/10.1063/1.4945570.
Tišma, M., Planinić, M., Bucić-Kojić, A., et al., 2018. Corn silage fungal-based solid-state pretreatment for enhanced biogas production in anaerobic co-digestion with cow manure. Bioresour. Technol. 253, 220–226. https://doi.org/10.1016/j.biortech.2018.01.037.
Travaini, R., Martín-Juárez, J., Lorenzo-Hernando, A., et al., 2016. Ozonolysis: an advantageous pretreatment for lignocellulosic biomass revisited. Bioresour. Technol. 199, 2–12. https://doi.org/10.1016/j.biortech.2015.08.143.
Vasco-Correa, J., Khanal, S., Manandhar, A., et al., 2018. Anaerobic digestion for bioenergy production: global status, environmental and techno-economic implications, and government policies. Bioresour. Technol. 247, 1015–1026. https://doi.org/10.1016/j.biortech.2017.09.004.
Venturin, B., Frumi Camargo, A., Scapini, T., et al., 2018. Effect of pretreatments on corn stalk chemical properties for biogas production purposes. Bioresour. Technol. 266, 116–124. https://doi.org/10.1016/j.biortech.2018.06.069.
Vivekanand, V., Olsen, E.F., Eijsink, V.G., et al., 2013. Effect of different steam explosion conditions on methane potential and enzymatic saccharification of birch. Bioresour. Technol. 127, 343–349. https://doi.org/10.1016/j.biortech.2012.09.118.
Wagner, A.O., Lackner, N., Mutschlechner, M., et al., 2018. Biological pretreatment strategies for second-generation lignocellulosic resources to enhance biogas production. Energies 11 (7), 1797. https://doi.org/10.3390/en11071797.
Wang, C., Shao, Z., Qiu, L., et al., 2021a. The solid-state physicochemical properties and biogas production of the anaerobic digestion of corn straw pretreated by microwave irradiation. RSC Adv. 11 (6), 3575–3584. https://doi.org/10.1039/d0ra09867a.
Wang, D., He, D., Liu, X., et al., 2019. The underlying mechanism of calcium peroxide pretreatment enhancing methane production from anaerobic digestion of waste activated sludge. Water Res. 164, 114934. https://doi.org/10.1016/j.watres.2019.114934.
Wang, J., Feng, K., Lou, Y., et al., 2021b. The synergistic effect of potassium ferrate and peroxymonosulfate application on biogas production and shaping microbial community during anaerobic co-digestion of a cow manure-cotton straw mixture. Bioresour. Technol. 333, 125166. https://doi.org/10.1016/j.biortech.2021.125166.
Wang, J., Lou, Y., Ma, D., et al., 2023. Co-treatment with free nitrous acid and calcium peroxide regulates microbiome and metabolic functions of acidogenesis and methanogenesis in sludge anaerobic digestion. Sci. Total Environ. 870, 161924. https://doi.org/10.1016/j.scitotenv.2023.161924.
Wang, S., Li, F., Zhang, P., et al., 2017. Ultrasound assisted alkaline pretreatment to enhance enzymatic saccharification of grass clipping. Energy Conv. Manag. 149, 409–415. https://doi.org/10.1016/j.enconman.2017.07.042.
Wang, S., Li, F., Wu, D., et al., 2018. Enzyme pretreatment enhancing biogas yield from corn stover: feasibility, optimization, and mechanism analysis. J. Agric. Food Chem. 66 (38), 10026–10032. https://doi.org/10.1021/acs.jafc.8b03086.

Witaszek, K., Pilarski, K., Niedbała, G., et al., 2020. Energy efficiency of comminution and extrusion of maize substrates subjected to methane fermentation. Energies 13 (8), 1887. https://doi.org/10.3390/en13081887.

Wu, J., Ein-Mozaffari, F., Upreti, S., 2013a. Effect of ozone pretreatment on hydrogen production from barley straw. Bioresour. Technol. 144, 344–349. https://doi.org/10.1016/j.biortech.2013.07.001.

Wu, J., Upreti, S., Ein-Mozaffari, F., 2013b. Ozone pretreatment of wheat straw for enhanced biohydrogen production. Int. J. Hydrogen Energy 38 (25), 10270–10276. https://doi.org/10.1016/j.ijhydene.2013.06.063.

Xiang, Y., Xiang, Y., Wang, L., 2017. Electron beam irradiation to enhance enzymatic saccharification of alkali soaked *Artemisia ordosica* used for production of biofuels. J. Environ. Chem. Eng. 5 (4), 4093–4100. https://doi.org/10.1016/j.jece.2017.07.058.

Ximenes, E., Kim, Y., Mosier, N., et al., 2010. Inhibition of cellulases by phenols. Enzyme Microb. Technol. 46 (3–4), 170–176. https://doi.org/10.1016/j.enzmictec.2009.11.001.

Xiong, J., Hassan, M., Wang, W., et al., 2020. Methane enhancement by the co-digestion of soybean straw and farm wastewater under different thermo-chemical pretreatments. Renew. Energy 145, 116–123. https://doi.org/10.1016/j.renene.2019.05.102.

Xu, H., Li, Y., Hua, D., et al., 2021. Effect of microaerobic microbial pretreatment on anaerobic digestion of a lignocellulosic substrate under controlled pH conditions. Bioresour. Technol. 328 https://doi.org/10.1016/j.biortech.2021.124852.

Xu, J., Zong, M.-H., Fu, S.-Y., et al., 2016. Correlation between physicochemical properties and enzymatic digestibility of rice straw pretreated with cholinium ionic liquids. ACS Sustain. Chem. Eng. 4 (8), 4340–4345. https://doi.org/10.1021/acssuschemeng.6b00860.

Yadav, M., Paritosh, K., Pareek, N., et al., 2019. Coupled treatment of lignocellulosic agricultural residues for augmented biomethanation. J. Clean. Prod. 213, 75–88. https://doi.org/10.1016/j.jclepro.2018.12.142.

Yang, J., Liu, X., Wang, D., et al., 2019. Mechanisms of peroxymonosulfate pretreatment enhancing production of short-chain fatty acids from waste activated sludge. Water Res. 148, 239–249. https://doi.org/10.1016/j.watres.2018.10.060.

Yang, J., Lan, X., Zhou, T., et al., 2023b. Effects of cold isostatic press pretreatment of rice straw on microstructure and efficiency of anaerobic digestion for methane production. Bioresour. Technol. 386, 129488. https://doi.org/10.1016/j.biortech.2023.129488.

Yang, J.-Y., Yu, Q.-F., Li, M.-F., 2023a. Freeze-thaw assisted maleic acid pretreatment of eucalyptus to prepare cellulose nanocrystals and degraded lignin. Bioresour. Technol. 384, 129365. https://doi.org/10.1016/j.biortech.2023.129365.

Yang, L., Xu, F., Ge, X., et al., 2015a. Challenges and strategies for solid-state anaerobic digestion of lignocellulosic biomass. Renew. Sustain. Energy Rev. 44, 824–834. https://doi.org/10.1016/j.rser.2015.01.002.

Yang, L., Li, X., Yuan, H., et al., 2023c. Enhancement of biomethane production and decomposition of physicochemical structure of corn straw by combined freezing-thawing and potassium hydroxide pretreatment. Energy 268, 126633. https://doi.org/10.1016/j.energy.2023.126633.

Yang, S.J., Yoo, H.Y., Choi, H.S., et al., 2015b. Enhancement of enzymatic digestibility of *Miscanthus* by electron beam irradiation and chemical combined treatments for bioethanol production. Chem. Eng. J. 275, 227–234. https://doi.org/10.1016/j.cej.2015.04.056.

Yu, Q., Zhuang, X., Yuan, Z., et al., 2010. Two-step liquid hot water pretreatment of *Eucalyptus grandis* to enhance sugar recovery and enzymatic digestibility of cellulose. Bioresour. Technol. 101 (13), 4895–4899. https://doi.org/10.1016/j.biortech.2009.11.051.

Yu, Q., Zhuang, X., Yuan, Z., et al., 2011. Step-change flow rate liquid hot water pretreatment of sweet sorghum bagasse for enhancement of total sugars recovery. Appl. Energy 88 (7), 2472–2479. https://doi.org/10.1016/j.apenergy.2011.01.031.

Yu, Q., Liu, R., Li, K., et al., 2019. A review of crop straw pretreatment methods for biogas production by anaerobic digestion in China. Renew. Sustain. Energy Rev. 107, 51–58. https://doi.org/10.1016/j.rser.2019.02.020.

Yuan, H., Lan, Y., Zhu, J., et al., 2019. Effect on anaerobic digestion performance of corn stover by freezing–thawing with ammonia pretreatment. Chin. J. Chem. Eng. 27 (1), 200–207. https://doi.org/10.1016/j.cjche.2018.04.021.

Yuan, X., Ma, L., Wen, B., et al., 2016. Enhancing anaerobic digestion of cotton stalk by pretreatment with a microbial consortium (MC1). Bioresour. Technol. 207, 293–301. https://doi.org/10.1016/j.biortech.2016.02.037.

Yuan, Z., Wen, Y., Li, G., 2018. Production of bioethanol and value added compounds from wheat straw through combined alkaline/alkaline-peroxide pretreatment. Bioresour. Technol. 259, 228–236. https://doi.org/10.1016/j.biortech.2018.03.044.

Zhang, K., Pei, Z., Wang, D., 2016. Organic solvent pretreatment of lignocellulosic biomass for biofuels and biochemicals: a review. Bioresour. Technol. 199, 21–33. https://doi.org/10.1016/j.biortech.2015.08.102.

Zhang, S., Keshwani, D.R., Xu, Y., et al., 2012. Alkali combined extrusion pretreatment of corn stover to enhance enzyme saccharification. Ind. Crops Prod. 37 (1), 352–357. https://doi.org/10.1016/j.indcrop.2011.12.001.

Zhang, Y., Chen, X., Gu, Y., et al., 2015. A physicochemical method for increasing methane production from rice straw: extrusion combined with alkali pretreatment. Appl. Energy 160, 39–48. https://doi.org/10.1016/j.apenergy.2015.09.011.

Zhao, B.-H., Chen, J., Yu, H.-Q., et al., 2017a. Optimization of microwave pretreatment of lignocellulosic waste for enhancing methane production: hyacinth as an example. Front. Env. Sci. Eng. 11 (6), 17. https://doi.org/10.1007/s11783-017-0965-z.

Zhao, C., Shao, Q., Ma, Z., et al., 2016. Physical and chemical characterizations of corn stalk resulting from hydrogen peroxide presoaking prior to ammonia fiber expansion pretreatment. Ind. Crops Prod. 83, 86–93. https://doi.org/10.1016/j.indcrop.2015.12.018.

Zhao, C., Qiao, X., Cao, Y., et al., 2017b. Application of hydrogen peroxide presoaking prior to ammonia fiber expansion pretreatment of energy crops. Fuel 205, 184–191. https://doi.org/10.1016/j.fuel.2017.05.073.

Zhao, C., Shao, Q., Chundawat, S.P.S., 2020. Recent advances on ammonia-based pretreatments of lignocellulosic biomass. Bioresour. Technol. 298, 122446. https://doi.org/10.1016/j.biortech.2019.122446.

Zhao, M.-J., Xu, Q.-Q., Li, G.-M., et al., 2019. Pretreatment of agricultural residues by supercritical CO_2 at 50-80 °C to enhance enzymatic hydrolysis. J. Energy Chem. 31, 39–45. https://doi.org/10.1016/j.jechem.2018.05.003.

Zhao, X., Cheng, K., Liu, D., 2009. Organosolv pretreatment of lignocellulosic biomass for enzymatic hydrolysis. Appl. Microbiol. Biotechnol. 82 (5), 815–827. https://doi.org/10.1007/s00253-009-1883-1.

Zheng, J., Rehmann, L., 2014. Extrusion pretreatment of lignocellulosic biomass: a review. Int. J. Mol. Sci. 15 (10), 18967–18984. https://doi.org/10.3390/ijms151018967.

Zheng, Y., Zhao, J., Xu, F., et al., 2014. Pretreatment of lignocellulosic biomass for enhanced biogas production. Prog. Energy Combust. Sci. 42, 35–53. https://doi.org/10.1016/j.pecs.2014.01.001.

Zhong, W., Zhang, Z., Luo, Y., et al., 2011. Effect of biological pretreatments in enhancing corn straw biogas production. Bioresour. Technol. 102 (24), 11177–11182. https://doi.org/10.1016/j.biortech.2011.09.077.

Zhou, J., Yan, B.H., Wang, Y., et al., 2016. Effect of steam explosion pretreatment on the anaerobic digestion of rice straw. RSC Adv. 6 (91), 88417–88425. https://doi.org/10.1039/c6ra15330e.

Zhou, Z., Ouyang, D., Liu, D., et al., 2023. Oxidative pretreatment of lignocellulosic biomass for enzymatic hydrolysis: progress and challenges. Bioresour. Technol. 367, 128208. https://doi.org/10.1016/j.biortech.2022.128208.

Zhou, Z.Y., Lei, F.H., Li, P.F., et al., 2018. Lignocellulosic biomass to biofuels and biochemicals: a comprehensive review with a focus on ethanol organosolv pretreatment technology. Biotechnol. Bioeng. 115 (11), 2683–2702. https://doi.org/10.1002/bit.26788.

Zhu, X., Liu, B., Sun, L., et al., 2023. Machine learning-assisted exploration for carbon neutrality potential of municipal sludge recycling via hydrothermal carbonization. Bioresour. Technol. 369, 128454. https://doi.org/10.1016/j.biortech.2022.128454.

Zhuang, X., Wang, W., Yu, Q., et al., 2016. Liquid hot water pretreatment of lignocellulosic biomass for bioethanol production accompanying with high valuable products. Bioresour. Technol. 199, 68–75. https://doi.org/10.1016/j.biortech.2015.08.051.

Zieliński, M., Dębowski, M., Kisielewska, M., et al., 2017. Comparison of ultrasonic and hydrothermal cavitation pretreatments of cattle manure mixed with straw wheat on fermentative biogas production. Waste Biomass Valoriz. 10 (4), 747–754. https://doi.org/10.1007/s12649-017-9977-y.

Zieliński, M., Dębowski, M., Kisielewska, M., et al., 2019. Cavitation-based pretreatment strategies to enhance biogas production in a small-scale agricultural biogas plant. Energy Sustain. Dev. 49, 21–26. https://doi.org/10.1016/j.esd.2018.12.007.

Zou, S., Wang, X., Chen, Y., et al., 2016. Enhancement of biogas production in anaerobic co-digestion by ultrasonic pretreatment. Energy Conv. Manag. 112, 226–235. https://doi.org/10.1016/j.enconman.2015.12.087.